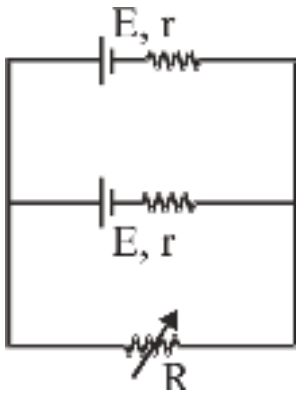
NOVA TG : @examlix

PHYSICS

1) The position of an object moving along the x-axis is given by: $x = A + Bt^2$ Where A = 10 m, B = 3 m/s^2, and t is measured in seconds.
(i) What is its velocity at **t = 3.0 s**?
(ii) What is the average velocity between **t = 1.0 s** and **t = 3.0 s**?

(1) Velocity = 18 m/s, Avg Velocity = 12 m/s
(2) Velocity = 9 m/s, Avg Velocity = 15 m/s
(3) Velocity = 18 m/s, Avg Velocity = 18 m/s
(4) Velocity = 12 m/s, Avg Velocity = 10 m/s

2) If E = 6V, r = 2Ω then maximum power dissipated through R is :-

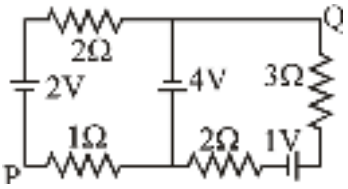


(1) 18 W
(2) 9 W
(3) 36 W
(4) 3 W

3) Water drops fall from a tap located on the roof of a building **16 meters** high at regular intervals of time. When the **fifth drop** is just about to leave the tap, the **first drop** touches the ground. Find the distance of the **second drop** from the **top (the tap)** at that instant. (Take **g = 10 m/s^2**)

(1) 4 meters
(2) 9 meters
(3) 12 meters
(4) 1 meter

4) In the circuit shown, what is the potential difference V_{PQ}?



(1) + 3V

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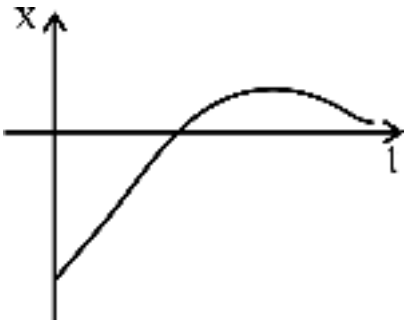
(2) + 2V

(3) +4V

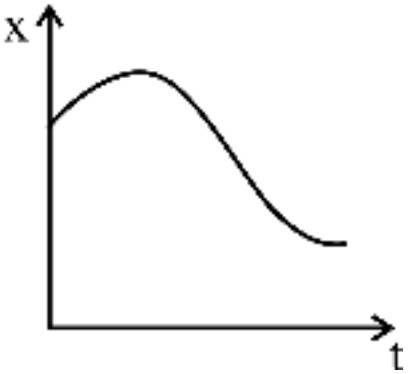
(4) +5V

5) Among the four graphs shown in figure, there is only one graph for which average velocity over the time interval (0, T) can vanish for a suitably chosen T. Which one is it ?

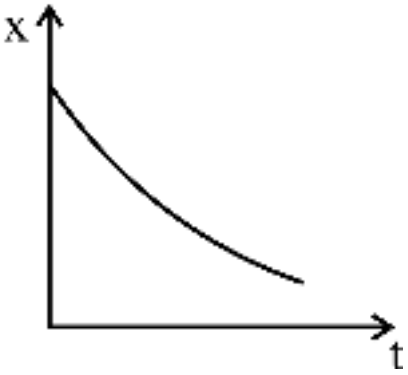
(1)



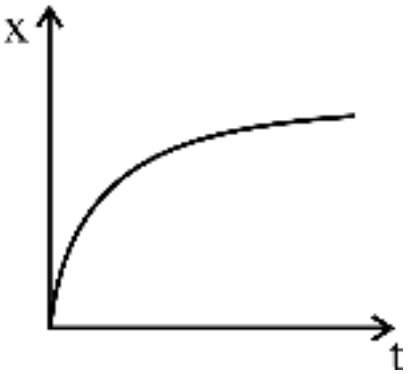
(2)



(3)

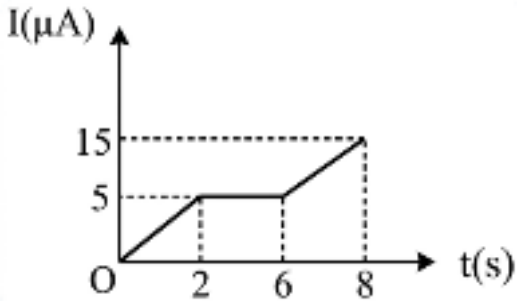


(4)



6) Following graph shows the variation of current with time. Find total charge flowing from t = 2s to t = 8s

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(1) 35 μC

(2) 40 μC

(3) 45 μC

(4) 20 μC

7) The charge flowing in a conductor varies with time as $Q = at - bt^2$. Then which of the following is not correct :-

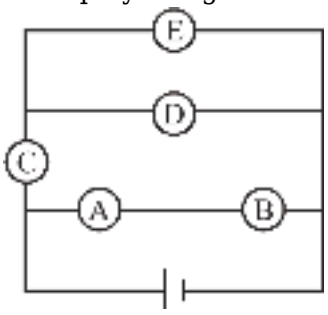
(1) The current decreases linearly with time.

(2) Current increases linearly with time

(3) Current falls to zero after time $t = \frac{a}{2b}$

(4) Current changes at a rate of 2b.

8) The given circuit is made up by using identical light bulbs. The light bulbs of maximum brightness of the following will be



(1) A

(2) C

(3) D

(4) E

9) A body is moving from rest under constant acceleration and let S_1 be the displacement in the first (p-1) sec and S_2 be the displacement in the first p sec. The displacement in $(p^2 - p + 1)^{th}$ sec. will be :-

(1) $S_1 + S_2$

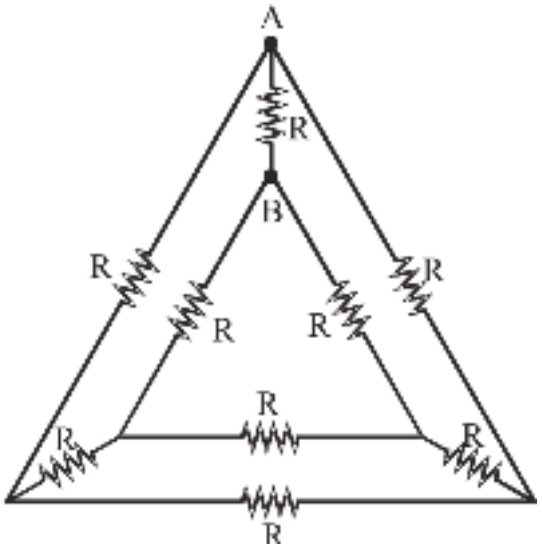
(2) $S_1 S_2$

(3) $S_1 - S_2$

(4) S_1 / S_2

10) Find equivalent resistance between the points A & B is :-

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(1) $\frac{3R}{5}$

(2) $\frac{2R}{5}$

(3) $\frac{R}{5}$

(4) $\frac{2R}{3}$

11) A car is moving with a velocity of 72 km/h. The driver suddenly applies brakes, which produce a uniform retardation of 5 m/s^2. Calculate : -

(i) The **time taken** by the car to come to a stop.

(ii) The **distance travelled** by the car before stopping (Stopping Distance).

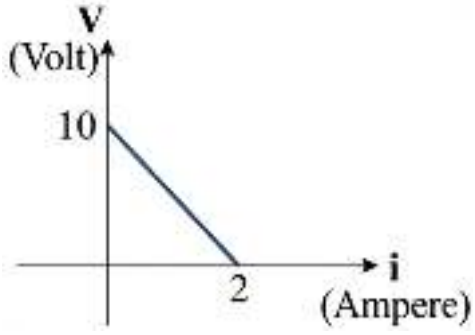
(1) Time = 4s, Distance = 40m

(2) Time = 2s, Distance = 20m

(3) Time = 4s, Distance = 20m

(4) Time = 5s, Distance = 50m

12) A battery of emf E and internal resistance r is connected across a resistance R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative.



(1) internal resistance of battery is 5Ω

(2) emf of the battery is 20V

(3) maximum current which can be taken from the battery is 4A

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(4) V-i graph can never be a straight line as shown in figure.

13) **Assertion :** In rectilinear motion when velocity is positive distance travelled increases and when velocity is negative distance travelled decreases.
Reason : Distance is length of the path covered by a particle.

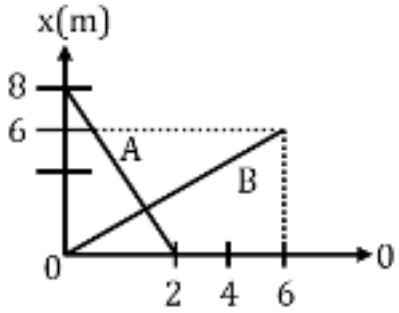
(1) If both assertion and reason are true and reason is the correct explanation of assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of assertion

(3) If assertion is true but reason is false.

(4) If assertion is false but reason is true.

14) The position - time graphs for two cars A and B moving on same track is shown in figure. The magnitude of their relative velocity is :



(1) 0

(2) 5m/s

(3) 3m/s

(4) 3.5 m/s

15) A particle **A** is projected vertically upwards from the ground with a velocity of **50 m/s**. At the same instant, another particle **B** is dropped from a tower of height **100 m** directly above it. At what distance from the **top of the tower** will the two particles collide? (Take **g = 10 m/s^2**)

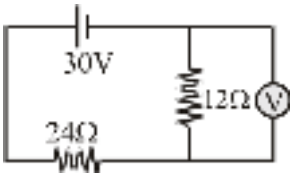
(1) 10 meters

(2) 20 meters

(3) 40 meters

(4) 50 meters

16) Voltmeter shown reads 6 V. The resistance of voltmeter is :-



(1) 75 Ω

(2) 12 Ω

(3) $\frac{800}{7}\Omega$

(4) 6 Ω

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17) A particle moves along a straight line such that its position (x) and time (t) are related by the equation: $t = 2x^2 + 3x$ Find the acceleration (a) of the particle in terms of its velocity (v).

(1) $a = -2v^2$

(2) $a = -4v^2$

(3) $a = -4v^3$

(4) $a = -8v^3$

18) Two wires have temperature coefficient α and –β and their resistance at 0°C temp are R_1 and R_2.

Now these two wires are connected in series. If $\frac{R_1}{R_2} > \frac{\beta}{\alpha}$ then :-

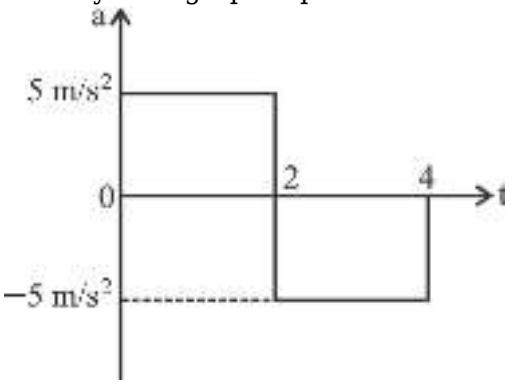
(1) Net resistance remains constant with respect to temperature

(2) Net resistance increases when temperature increases

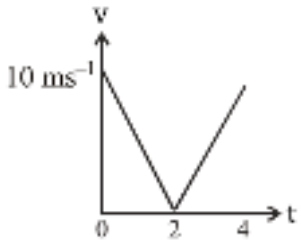
(3) Net resistance increases when temperature decreases

(4) None

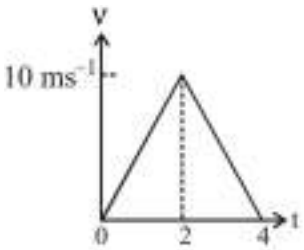
19) A body starts from rest at t = 0 and undergoes an acceleration as shown in the graph. Which velocity time graph represent the motion of the body in 4 second ?



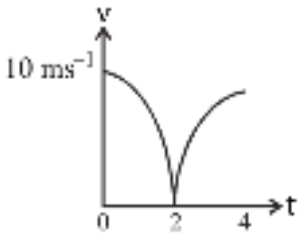
(1)



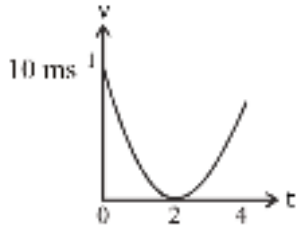
(2)



(3)



Watermarkly

(4) 

20) A coin is released inside a lift at a height of 2m from the floor of the lift. The height of the lift is 10m. The lift is moving with an acceleration of 9 m/s^2 downwards. The time after which the coin will strike with the floor lift is ($g=10m/s^2$)

(1) 4s

(2) 2s

(3) $\frac{4}{\sqrt{21}}s$

(4) $\frac{2}{\sqrt{21}}s$

21) Select the incorrect statement :

(1) For uniform motion, acceleration is zero and x-t graph is a straight line.

(2) The zero velocity of a particle at any instant does not necessarily imply zero acceleration at that instant.

(3) If particle is moving with constant velocity then |average velocity| = average speed.

(4) A particle in one-dimensional motion with positive value of acceleration must be speeding up.

22) A constant electric current I is passed through a straight conductor of length □. If S is specific charge of electron then the total momentum of electrons is :-

(1) $\frac{IS}{\ell}$

(2) $\frac{I\ell}{S}$

(3) $\frac{S\ell}{I}$

(4) $\frac{2I\ell}{S}$

23) A particle is dropped from a hot air balloon. Balloon is rising up with uniform speed 10 m/s. If particle reaches to ground in 7s. Find out height from where particle is dropped.

(1) 100 m

(2) 125 m

(3) 150 m

(4) 175 m

24) A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H. If the ball attains maximum height in 2 s and lands on the building in 3 s after launch, then value of H is _____ m. ($g = 10 m/s^2$)

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(1) 20

(2) 10

(3) 25

(4) 15

25) The position vector of a particle is given as $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j}$. Find the time after which the velocity vector and acceleration vector becomes perpendicular to each other.

(1) 2 sec

(2) 1 sec

(3) 1.5 sec

(4) 5 sec

26) An electric kettle has two heating coils. When one coil is used, water in the kettle boils in 5 minutes, while when second coil is used, same water boils in 10 minutes. If the two coils, connected in parallel are used simultaneously, the same water will boil in time :-

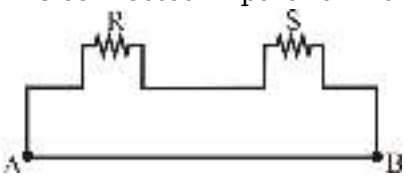
(1) 3 min 20 sec

(2) 3 min 33 sec

(3) 7 min 30 sec

(4) 2 min 30 sec

27) In a meter bridge, the null point is found at a distance of 25 cm from A. If now a resistance of 10 Ω is connected in parallel with S, the null point occurs at mid point of AB. The value of R is :-



(1) 6.67 Ω

(2) 1.67 Ω

(3) 2.67 Ω

(4) 4.67 Ω

28) A body is projected vertically upwards if $t_1 = 2$ sec and $t_2 = 8$ sec be the times at which it is at height h above the projection while ascending and descending respectively then h is-

(1) 40 m

(2) 80 m

(3) 100 m

(4) 160 m

29) Two conductors made of same material have length L and 2L but have equal resistance. The two are connected in series in a circuit then what will be ratio of drift velocity of free electron in two conductors :-

(1) 1 : 2

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(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

30) A particle is projected with $\vec{u} = 30\hat{i} + 40\hat{j}$ m/s from ground. Then match the two columns (in respective units)

	Column-I		**Column-II**
(A)	Time of flight	P.	30
(B)	Maximum height	Q.	80
(C)	Range	R.	240
(D)	Horizontal velocity at t = 3s.	S.	8
		T.	10

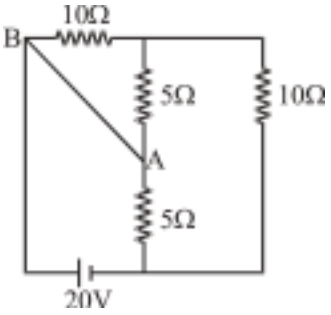
(1) A-S, B-Q, C-R, D-P

(2) A-P, B-Q, C-R, D-S

(3) A-S, B-R, C-Q, D-P

(4) A-T, B-Q, C-S, D-P

31) In the circuit shown find the current in branch AB of the circuit.



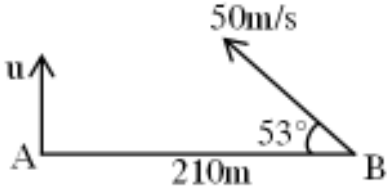
(1) 5 A

(2) 0.5 A

(3) $\frac{11}{3}$A

(4) Zero

32) Find out collision time of projectile A & B in given figure. When they are projected simultaneously from ground -



(1) 5 s

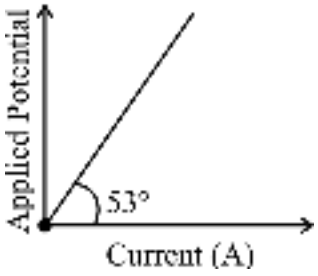
(2) 6 s

(3) 7 s

(4) 8 s

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33) The variation of applied potential and current flowing through a given wire is as shown in figure. The length of the wire is 50 cm and diameter of wire is measured to be 0.2 cm. The resistivity of the wire, is given near to :



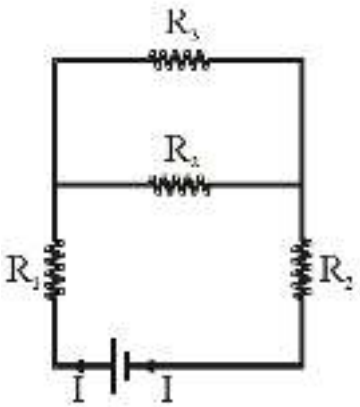
(1) $2 \times 10^{-4}\ \Omega^{m}$

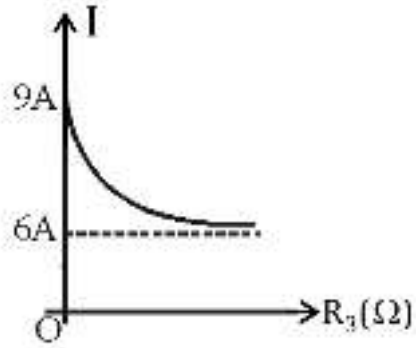
(2) $4 \times 10^{-6}\ \Omega^{m}$

(3) $8 \times 10^{-6}\ \Omega^{m}$

(4) $4 \times 10^{-4}\ \Omega^{m}$

34) In the circuit shown R_3 is a variable resistance. As value of R_3 is changed, current I through the cell varies as shown in graph, then value of R_1 and R_2 are, respectively :-





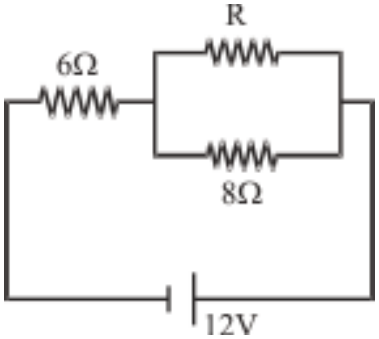
(1) 4Ω, 2Ω

(2) 2Ω, 4Ω

(3) 2Ω, 2Ω

(4) 1Ω, 4Ω

35) In the circuit shown in figure, the power which is dissipated as heat in the 6Ω resistor is 6W. What is the value of resistance R in the circuit?



(1) 6 Ω

(2) 10 Ω

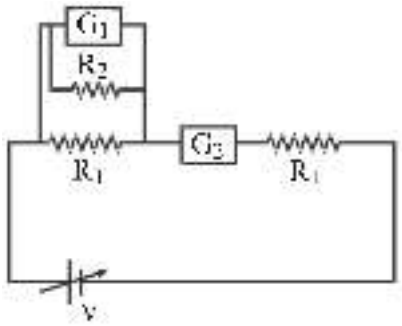
(3) 13 Ω

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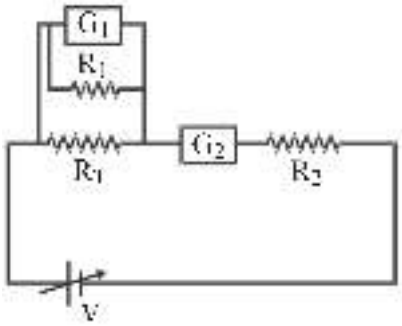
(4) 24 Ω

36) To verify Ohm's law, a student is provided with a test resistor R_T, a high resistance R_1, a small resistance R_2, two identical galvanometers G_1 and G_2, and a variable voltage source V. The correct circuit to carry out the experiment is :-

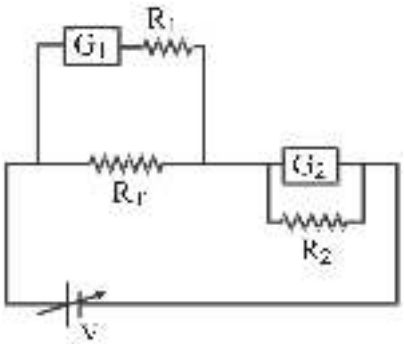
(1)



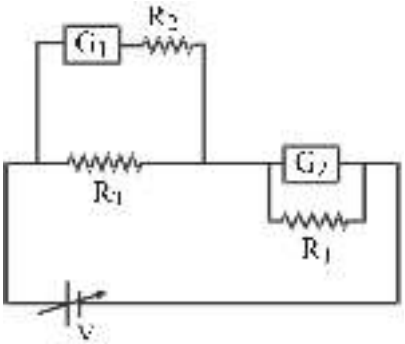
(2)



(3)



(4)



37) An aeroplane flying horizontally at a speed of 500 m/s releases a bomb. The bomb strikes the ground in 10 s. The angle of velocity from horizontal at which the bomb strikes the ground is :- (g = 10 m/s^2) :

(1) $\tan^{-1}$ (1/5)

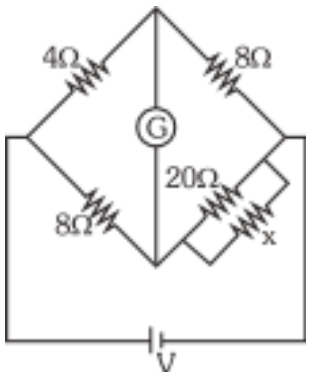
(2) 60°

(3) 45°

Watermarkly

(4) $\tan^{-1}(5)$

38) Find the value of x for which bridge is balanced



(1) 80 Ω

(2) 20 Ω

(3) 160 Ω

(4) 16 Ω

39) A galvanometer has a coil of resistance 100Ω showing a full–scale deflection at 50 μA. Consider following statements.
(A) The resistance needed to use it as a voltmeter of range 50 V is 10^6 Ω
(B) The resistance needed to use it as a voltmeter of range 50 V is 10^5 Ω
(C) The resistance needed to use it as an ammeter of range 10 mA is 0.5 Ω
(D) The resistance needed to use it as an ammeter of range 10 mA is 1.0 Ω
Select correct alternative :-

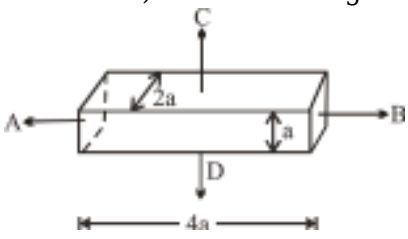
(1) Only A, D

(2) Only A, C

(3) Only B, D

(4) Only B, C

40) A conductor with rectangular cross section has dimensions (a × 2a × 4a) as shown in figure.



Resistance across AB is x, across CD is y. Then the value of $\frac{x}{y}$ is :-

(1) 2

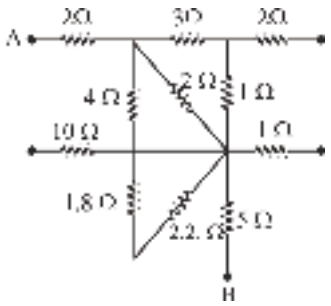
(2) 4

(3) 8

(4) 16

41) What is the equivalent resistance between the points A and B of the network :-

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(1) $\frac{57}{7}\Omega$

(2) 8 Ω

(3) 6 Ω

(4) $\frac{57}{5}\Omega$

42) A swimmer wishes to cross a 500m wide river flowing at 5 km/h. His speed with respect to water is 3 km/hr. (a) If he heads in a direction making an angle θ with the flow, find the time he takes to cross the river.
(b) Find the shortest possible time to cross the river

(1) $t = \frac{0.5}{3\cos\theta}h$, t_{min} = 15 min

(2) $t = \frac{0.5}{3\sin\theta}h$, t_{min} = 10 min

(3) $t = \frac{0.5}{5\sin\theta}h$, t_{min} = 6 min

(4) $t = \frac{0.5}{3\sin\theta}h$, t_{min} = 20 min

43) n identical cells are joined in series with its two cells A and B in the loop with reversed polarities. EMF of each cell is E and internal resistance r. Potential difference across cell A or B is (here n > 4) :-

(1) $\frac{2E}{n}$

(2) $2E\left(1-\frac{1}{n}\right)$

(3) $\frac{4E}{n}$

(4) $2E\left(1-\frac{2}{n}\right)$

44) A bullet fired into a fixed target loses half of its velocity after penetrating 3 cm. How much further it will penetrate before coming to rest? [Assume constant retardation]

(1) 1.5 cm

(2) 1.0 cm

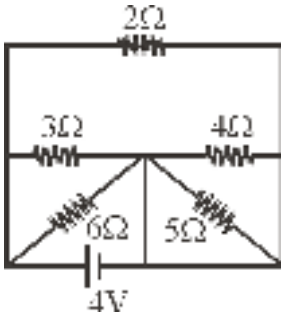
(3) 3.0 cm

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(4) 2.0 cm

45)

Find current through 4V battery



(1) 1A

(2) 2A

(3) 3A

(4) 4A

CHEMISTRY

1) For the reaction $2A \rightarrow B + 3C$; if – $\frac{d[A]}{dt} = k_1[A]^2$; $\frac{d[B]}{dt} = k_2[A]^2$; $\frac{d[C]}{dt} = k_3[A]^2$ the correct relation between k_1, k_2 and k_3 is :

(1) $k_1 = k_2 = k_3$

(2) $2k_1 = k_2 = 3k_3$

(3) $4k_1 = k_2 = 3k_3$

(4) $\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

2) The rate constant of nth order has units :

(1) $\text{litre}^{1-n}\ \text{mol}^{1-n}\ \text{sec}^{-1}$

(2) $\text{Mol}^{1-n}\ \text{litre}^{1-n}\ \text{sec}$

(3) $\text{Mol}^{1-n^2}\text{litre}^{n^2}\text{sec}^{-1}$

(4) $\text{Mol}^{1-n}\ \text{litre}^{n-1}\ \text{sec}^{-1}$

3) The rate constant of a reaction depends on

(1) temperature

(2) pressure

(3) extent of reaction

(4) initial concentration of the reactant

4) The molecularity of a complex reaction given below is :

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$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

(1) 1

(2) 2

(3) 3

(4) has no meaning

5) $A_2 + B_2 \rightarrow 2AB$; R. O. R. = $k[A_2]^a [B_2]^b$

Initial $[A_2]$	**Initial $[B_2]$**	**R.O.R. (r) Ms^{-1}**
0.2	0.2	0.04
0.1	0.4	0.04
0.2	0.4	0.08

Order of reaction with respect to A_2 and B_2 are respectively:

(1) a = 1, b = 1

(2) a = 2, b = 0

(3) a = 2, b = 1

(4) None

6) Reaction A → B follows second order kinetics. Doubling the concentration of A will increase the rate of formation of B by a factor of :

(1) 1/4

(2) 1/2

(3) 2

(4) 4

7) If 'a' is the initial concentration of the reactant, the half-life period of the reaction of nth order is inversely proportional to :

(1) a^{n-1}

(2) a^n

(3) a^{1-n}

(4) a^{n+1}

8) The units of rate of reaction and rate constant are same for a :

(1) zero order reaction

(2) first order reaction

(3) second order reaction

(4) third order reaction

9) $CH_3COOC_2H_5(aq) + H_2O(\square) \xrightarrow{H^+(aq)} CH_3COOH(aq) + C_2H_5OH(aq)$. What type of reaction is this?

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(1) Unimolecular elementary

(2) Pseudo first order

(3) Zero order

(4) Second order

10) For an elementary reaction $2A + B \rightarrow A_2B$ if the volume of vessel is quickly reduced to half of its original volume then rate of reaction will

(1) remain unchanged

(2) increase four times

(3) increase eight times

(4) decrease eight times

11) A first order reaction is 75% completed in 100 minutes. How long time will it take for its 87.5% completion?

(1) 125 min

(2) 150 min

(3) 175 min

(4) 200 min

12) The rate constant for a first order reaction whose half life is 480 sec, is:

(1) 1.44×10^{-3} sec^{-1}

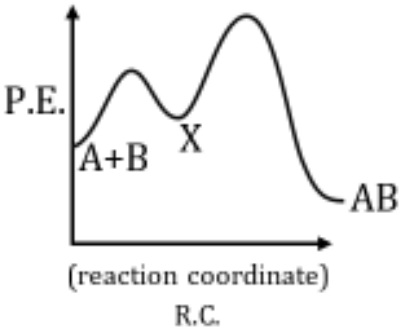
(2) 1.44×10^{-4} sec^{-1}

(3) 0.72×10^{-3} sec^{-1}

(4) 2.88×10^{-3} sec^{-1}

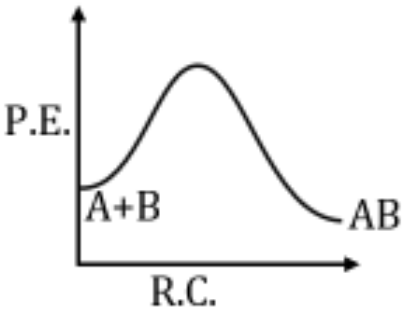
13) For an exothermic chemical process occurring in two steps as follows
(i) $A + B \rightarrow X$ (slow) (ii) $X \rightarrow AB$ (fast)
The progress of reaction can be best described by:

(1)

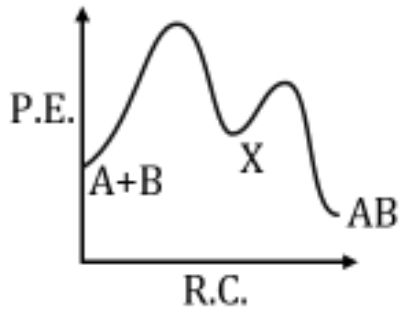


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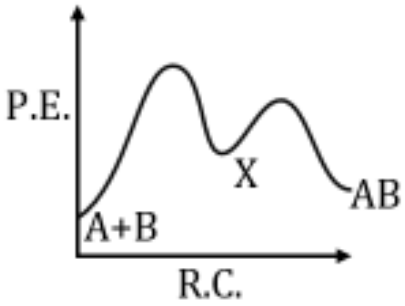
(2)



(3)



(4)



14) If the value of equilibrium constant for a particular reaction is 1.6×10^{12}, then at equilibrium, the system will contain

(1) mostly products

(2) similar amounts of reactants and products

(3) all reactants

(4) mostly reactants

15) If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K,

the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be

(1) $\frac{1}{2}K$

(2) K

(3) K^2

(4) $K^{1/2}$

16) In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be

(1) Reduced to half its original value

(2) Reduced to one fourth of its original value

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(3) Doubled

(4) Constant

17)

The equilibrium constant for the reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 2000 K. In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant, in presence of the catalyst, at 2000 K is:

(1) 40×10^{-4}

(2) 4×10^{-4}

(3) 4×10^{-3}

(4) difficult to compute without more data.

18) Partial pressure of O_2 in the reaction $2Ag_2O(s) \rightleftharpoons 4\,Ag(s) + O_2(g)$ is

(1) K_p

(2) $\sqrt{K_p}$

(3) $\sqrt[3]{K_p}$

(4) $2K_p$

19) A quantity of PCl_5 was heated in a 10 dm^3 vessel at 250°C : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium, the vessel contains 0.1 mole of PCl_5 and 0.2 mole of Cl_2. The equilibrium constant of the reaction is

(1) 0.04

(2) 0.025

(3) 0.02

(4) 0.05

20) $A + B \rightleftharpoons C + D$. If initially the concentrations of A and B are equal but at equilibrium, concentration of D will be twice that of A, then what will be the equilibrium constant of the reaction ?

(1) $\frac{4}{9}$

(2) $\frac{9}{4}$

(3) $\frac{1}{9}$

(4) 4

21) Which of the following is true-

(1) Active mass of pure liquids and solids are taken unity.

(2) Active mass of gases can be expressed is terms of atm.

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(3) Law of mass action is applicable on reversible reaction

(4) All of above

22) The equilibrium constant (K_c) for the reaction $HA + B \rightleftharpoons BH^+ + A^-$ is 100. If the rate constant for the forward reaction is 10^5, then rate constant for the backward reaction is

(1) 10^7

(2) 10^3

(3) 10^{-3}

(4) 10^{-5}

23) **Statement-1 :-** In exothermic reaction, heat is evolved during reaction.
Statement-2 :- Product is more stable then reactant.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

24) The aqueous solution of following salt will have the lowest pH

(1) NaClO

(2) $NaClO_4$

(3) $NaClO_3$

(4) $NaClO_2$

25) pK_a of weak acid (HA) and pK_b of weak base (BOH) are 3.2 and 3.4 respectively. The pH of their salt (BA) solution is

(1) 7.0

(2) 1.0

(3) 7.2

(4) 6.9

26) The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is

(1) 8.50

(2) 5.50

(3) 7.75

(4) 6.25

27)

Match the entries of column I with appropriate entries of column II and choose the correct option.

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	Column-I		Column-II
(A)	pK_b of X^- (K_a of HX = 10^{-6})	(p)	6.96
(B)	pH of 10^{-8} M HCl	(q)	8
(C)	pH of 10^{-2} M acetic acid solution (Take K_a of acetic acid = 2×10^{-5})	(r)	3.3
(D)	pH of a solution obtained by mixing equal volumes of solutions with pH 3 and 5.	(s)	3.35

(1) A-q, B-p, C-r, D-s

(2) A-r, B-s, C-q, D-p

(3) A-q, B-p, C-s, D-r

(4) A-p, B-q, C-s, D-r

28) What will be the pH of pure water at 60°C ?

(1) Equal to 7.0

(2) Greater than 7.0

(3) Less than 7.0

(4) Equal to zero

29) The K_a values of acetic acid, hypochlorous acid and formic acid are 1.74×10^{-5}, 3.0×10^{-8} and 1.8×10^{-4} respectively Which of the following orders of pH of 0.1 mol dm^{-3} solutions of these acids is correct ?

(1) acetic acid > hypochlorous acid > formic acid

(2) hypochlorous acid > acetic acid > formic acid

(3) formic and > hypochlorous acid > acetic acid

(4) formic acid > acetic acid > hypochlorous acid

30) K_a for CH_3COOH is 1.8×10^{-5} and K_b for NH_4OH is 1.8×10^{-5}. The pH of ammonium acetate will be

(1) 7.005

(2) 4.75

(3) 7.0

(4) Between 6 and 7

31) **Assertion (A):** The ionisation of hydrogen sulphite in water is low in the presence of hydrochloric acid.
Reason (R): Hydrogen sulphite is a weak acid.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false

(4) Both A and R are false

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32) **Assertion (A):** Aqueous solution of ammonium carbonate is basic.
Reason (R): Acidic/basic nature of a salt solution of a salt of weak acid and weak base depends on K_a and K_b value of the acid and the base forming it.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false.

33) At 25°C, the dissociation constant of a base, BOH is 1.0×10^{-12}. The concentration of hydroxyl ions in 0.01 M aqueous solution of the base would be:

(1) 1.0×10^{-6} mol L^{-1}
(2) 1.0×10^{-7} mol L^{-1}
(3) 2.0×10^{-6} mol L^{-1}
(4) 1.0×10^{-5} mol L^{-1}

34) Aqueous solution of which of the following compounds is the best conductor of electric current ?

(1) Ammonia, NH_3
(2) Fructose, $C_6H_{12}O_6$
(3) Acetic acid, $C_2H_4O_2$
(4) Hydrochloric acid HCl

35) What is the conjugate base of OH^- ?

(1) O_2
(2) H_2O
(3) O^-
(4) O^{2-}

36) pH value of which of the following is NOT equal to one?

(1) 0.1 M CH_3COOH
(2) 0.1 M HNO_3
(3) 0.05 M H_2SO_4
(4) 50 cm^3, 0.4M HCl + 50 cm^3, 0.2 M NaOH

37) An acid HA ionises as $HA \rightleftharpoons H^+ + A^-$
The pH of 1.0 M solution is 5. Its dissociation constant would be

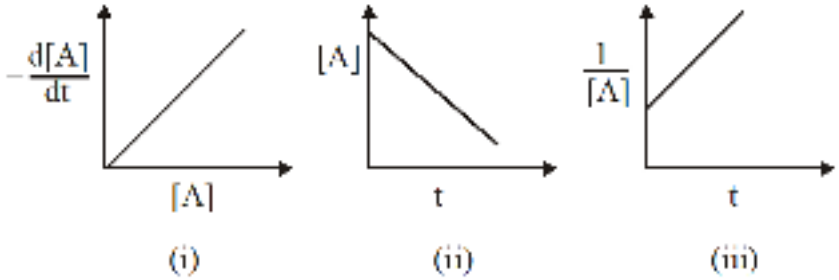
(1) 1×10^{-10}
(2) 5
(3) 5×10^{-8}
(4) 1×10^{-5}

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38) The pH of a 0.1 M aqueous solution of a weak acid (HA) is 3. What is its degree of dissociation?

(1) 1%
(2) 10%
(3) 50%
(4) 25%

39) Consider the plots for the types of reaction
$nA \rightarrow B + C$



These plots respectively correspond to the reaction orders:

(1) 0, 2, 1
(2) 0, 1, 2
(3) 1, 1, 2
(4) 1, 0, 2

40) The acidic hydrolysis of ester is –

(1) Psuedo I^{st} order reaction
(2) IInd order reaction
(3) Trimolecular reaction
(4) All of the above

41) For a first order reaction, how much half lives are nearly required for 99.9% completion of reaction?

(1) 2
(2) 6
(3) 8
(4) 10

42) $xA + yB \longrightarrow zC$.
If $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{3}{2}\frac{d[C]}{dt}$, then x, y and z are:

(1) 1, 1, 1
(2) 3, 2, 3
(3) 3, 3, 2
(4) 2, 2, 3

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43) A and B react as per the reaction
$A(g) + 2B(g) \rightarrow C(g)$
From the data given below identify the Rate Law

[A] (M)	[B] (M)	ROR M sec^{-1}
0.3	0.4	2×10^{-3}
0.6	0.8	8×10^{-3}
0.6	0.4	4×10^{-3}

(1) $R = K [A]^2 [B]^1$

(2) $R = K [A] [B]$

(3) $R = K [A]^1 [B]^0$

(4) $R = K [A] [B]^{-1}$

44) Use the data to determine the orders of A and B in the reaction ; A + B → products

Experiment	[A], mol L^{-1}	[B], mol L^{-1}	Rate, mol L^{-1} s^{-1}
1	0.10	0.10	0.0090
2	0.20	0.10	0.036
3	0.10	0.20	0.018

(1) Rate = k [A] [B]

(2) Rate = k $[A]^2$ [B]

(3) Rate = k [A] $[B]^2$

(4) Rate = k $[A]^2$ $[B]^2$

45) Consider the Arrhenius equation given below and mark the **CORRECT** option, $k = Ae^{-E_a/RT}$

(1) Rate constant increases exponentially with increasing activation energy and decreasing temperature.

(2) Rate constant remains constant with increasing activation energy and decreasing temperature.

(3) Rate constant increases exponentially with decreasing activation energy and decreasing temperature.

(4) Rate constant increases exponentially with decreasing activation energy and increasing temperature.

BIOLOGY

1) In water hyacinth and water lily, pollination takes place by:

(1) Water current only

(2) Wind and water

(3) Insects and wind

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(4) Insects and water

2) Which of the following is not a part of the inner most whorl of the flower ?

(1) Stamen
(2) Style
(3) Ovary
(4) Stigma

3) How many plants in the list given below are polycarpic? Wheat, rice, maize, bamboo, mango, apple. guava, lemon, orange

(1) Six
(2) Three
(3) Four
(4) Five

4) The pollen tube usually enters the embryo sac

(1) Through one of the synergids
(2) By directly penetrating the egg
(3) Between one synergid and central cell
(4) By knocking off the antipodal cells

5) One of the male gamete moves towards the egg cell & fuses with it, the process is known as -

(1) Syngamy
(2) Triple fusion
(3) Double fertilization
(4) Autogamy

6) In the given statements (A-D), how many is/are correct?
A. The portion of the embryonal axis above the level of attachment of scutellum is the hypocotyl.
B. In some seeds, remnants of nucellus are persistent as in black pepper and beet.
C. Most of monocot seeds are endospermic seeds.
D. Micropyle facilitates entry of pollen tube into the seed during germination.

(1) Four
(2) Three
(3) Two
(4) One

7) Dicot embryo develops at the micropylar end of the embryo sac in which of the following sequences?

(1) Zygote → Pro-embryo → Mature embryo → Globular embryo
(2) Zygote → Pro-embryo → Globular embryo → Heart shaped embryo→ Mature embryo

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(3) Zygote → Globular embryo→ Pro-embryo → Heart shaped embryo → Mature embryo
(4) Zygote → Pro-embryo → Heart shaped embryo → Globular embryo → Mature embryo

8) Plant which prevent both autogamy and geitonogamy ?

(1) Cucumber
(2) Papaya
(3) Castor
(4) Maize

9) **Statement-I :-** In majority of angiosperms anther is bilobed, with each lobe having two theca.
Statement-II :- Cleistogamous flowers produce assured seed set even in the absence of pollinators.

(1) Both statement I and statement II are correct
(2) Statement I is correct and statement II is incorrect
(3) Statement I is incorrect and statement II is correct
(4) Both statement I and II are incorrect

10) Identify the incorrect statement related to pollination :-

(1) Pollination by wind is more common amongst abiotic pollination
(2) Flowers produce foul odours to attract flies and beetles to get pollinated
(3) Moths and butterflies are the most dominant pollinating agents among insects
(4) Pollination by water is quite rare in flowering plants

11) **Assertion :-** Cleistogamous flowers are invariably autogamous.
Reason :- These flowers do not open at all.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion
(3) Assertion is true but the reason is false
(4) Both assertion & reason are false

12) The ploidy levels of the cells of the secondary nucleus, PMC, the functional megaspore and female gametophyte are :

(1) 3N, N, 2N, N
(2) N, N, 2N, N
(3) 2N, 2N, N, N
(4) N, 2N, 2N, N

13) The female gametophyte/embryo sac of a typical dicot embryo sac is :

(1) 6-celled and 7-nucleate
(2) 8-celled and 8-nucleate
(3) 7-celled and 8-nucleate

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(4) 1-celled and 7-nucleate

14) Identify the incorrect statement related to pollination :

(1) Majority of plants use biotic agents for pollination
(2) Wind pollination is quite common in grasses
(3) Wind and water pollinated flowers are not very colourful and do not produce nectar
(4) *Vallisnaria* and *Hydrilla* are pollinated by insects or wind.

15) Which of the following is not an adaptation for anemophily ?

(1) Presence of well exposed stamens
(2) Often presence of single ovule in an ovary
(3) Sticky pollen grain
(4) Large and often feathery stigma

16) What is incorrect in the following statements?
(a) Microsporangium is near circular in outline
(b) Four wall layers of microsporangium help in dehiscence of anther
(c) Vegetative cell nucleus of pollen grain is irregular in shape
(d) Pollen grains are about 25-50 Å in diameter

(1) only a
(2) only b, c
(3) All
(4) only b, d

17) In angiosperms, during double fertilization, one male gamete fuses with the egg cell and the other fuses with the central cell. What is the ploidy (chromosome number) of the resulting structures respectively?

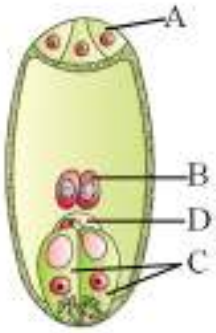
(1) Haploid (n) and Diploid (2n)
(2) Diploid (2n) and Triploid (3n)
(3) Diploid (2n) and Haploid (n)
(4) Triploid (3n) and Diploid (2n)

18) Find the correct statements from the given statements
(a) Early stages of embryogeny are similar in maize and pea
(b) Hypocotyl terminates at lower end into root tip
(c) Most zygotes divide after complete endosperm formation
(d) Part of embryonal axis located just below scutellum is called epicotyl

(1) a, b, c
(2) b, c
(3) a, b, d
(4) a, b

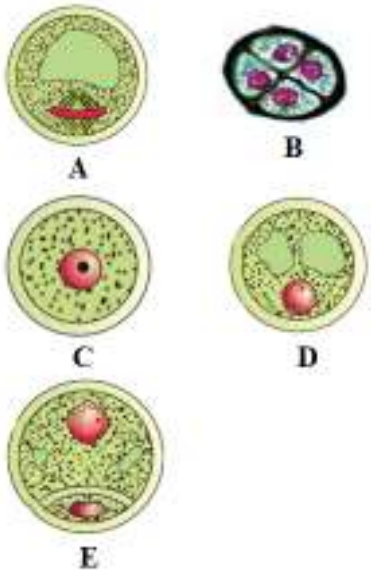
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19) Study the following diagram carefully and identify the structure which is involved in triple fusion



(1) A: Synergids
(2) B: Polar nuclei
(3) C: Egg cell
(4) D: Polar nuclei

20) Few events regarding pollen development is given in figures. Arrange the figures in correct sequence and select the correct option from options given below :-



(1) B, D, C, A, E
(2) B, C, A, D, E
(3) B, C, D, A, E
(4) B, A, C, D, E

21) Which of the following is true regarding pollen viability?

(1) Pollen viability is constant in all member of one family.
(2) Pollen viability mainly depends on temperature but not on humidity.
(3) In some members of Solanaceae family pollen remain viable for months.
(4) In wheat and rice pollen remain viable for many months.

22) Match **List-I** with **List-II.**

	List-I		List-II

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A.	Scutellum	I.	Persistent nucellus
B.	Non-albuminous seed	II.	Cotyledon of Monocot seed
C.	Epiblast	III.	Groundnut
D.	Perisperm	IV.	Rudimentary cotyledon

Choose the option with all **correct** matches.

(1) A–II, B–III, C–IV, D–I

(2) A–IV, B–III, C–II, D–I

(3) A–IV, B–III, C–I, D–II

(4) A–II, B–IV, C–III, D–I

23)

(a)	Exine	(i)	Large irregular nucleus
(b)	Intine	(ii)	Sporopollenin absent
(c)	Germ pore	(iii)	Sporopollenin
(d)	Vegetative cell	(iv)	Pecto-Cellulose

Choose the correct match :-

(1) a–iii, b–iv, c–i, d–ii

(2) a–iv, b–ii, c–iii, d–i

(3) a–iii, b–iv, c–ii, d–i

(4) a–i, b–ii, c–iv, d–iii

24) Match the **Column I** with **Column II** :-

Column I		**Column II**	
A.	Funicle	I.	Mass of cells within ovule with more food
B.	Hilum	II.	Basal part of ovule
C.	Integument	III.	One or two protective layers of ovule
D.	Chalaza	IV.	Region where body of ovule fuses with funicle
E.	Nucellus	V.	Stalk of ovule

(1) A-I, B-II, C-III, D-IV, E-V

(2) A-V, B-IV, C-III, D-II, E-I

(3) A-IV, B-II, C-I, D-III, E-V

(4) A-I, B-III, C-V, D-II, E-IV

25) **Assertion (A) :-** Endosperm is persistant in castor
Reason (R) :- Endosperm is not consumed completely in castor during embryo formation.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is incorrect but **(R)** is not correct.

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(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

26) Monoecious plant with unisexual flowers prevents :-

(1) Autogamy
(2) Geitonogamy
(3) Both autogamy and geitonogamy
(4) Pollination is not possible

27) Identify the wrong statement regarding post fertilization development.

(1) The ovary wall develops into pericarp.
(2) The outer integument of the ovule develops into tegmen.
(3) The triple fusion nucleus develops into endosperm.
(4) The ovule develops into seed.

28) Embryo axis above the cotyledons is callled as :

(1) Hypocotyl
(2) Epicotyl
(3) Raphe
(4) Funicle

29) Select the false statement :-

(1) Endosperm development precedes embryo development
(2) The primary endosperm cell divides repeatedly and forms a triploid endosperm tissue
(3) The cells of endosperm tissue are filled with reserve food materials and are used for the nutrition of the developing embryo
(4) Embryo development precedes endosperm development

30) The pollen grain represents the :-

(1) Female gametophyte
(2) Male gamete
(3) Male gametophyte
(4) Female gamete

31) A male patient has dysfunctional interstitial (Leydig) cells due to injury. Which process will be most directly and severely affected?

(1) Seminal fluid production
(2) Androgen secretion required for spermatogenesis and secondary sexual characters
(3) Ovulation in partner
(4) Uterine contraction during labour

32) If FSH secretion is experimentally blocked after primary follicle formation in a female, the most

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immediate effect will be:

(1) Immediate ovulation
(2) Arrest of follicular growth at primary stage; no Graafian follicle formation
(3) Premature corpus luteum degeneration
(4) Increased LH surge

33) The secondary oocyte is released at ovulation arrested at which meiotic stage?

(1) Metaphase-I
(2) Metaphase-II
(3) Anaphase-I
(4) Telophase-II

34) A sperm without acrosome reaches the ovum but cannot fertilise it. The reason is:

(1) It cannot penetrate corona radiata and zona pellucida
(2) It cannot swim in female tract
(3) Cortical reaction fails
(4) Implantation is blocked

35) During early pregnancy, the corpus luteum is maintained by:

(1) Progesterone alone
(2) Human chorionic gonadotropin (hCG) secreted by trophoblast
(3) Estrogen from ovary
(4) Relaxin

36) Trophoblast cells of blastocyst primarily contribute to:

(1) Embryo proper
(2) Placenta formation by forming chorionic villi
(3) Amniotic sac
(4) Yolk sac

37) Milk ejection (let-down reflex) is caused by:

(1) Prolactin
(2) Oxytocin acting on myoepithelial cells
(3) Estrogen
(4) Progesterone

38) In the absence of fertilisation, corpus luteum degenerates mainly because of:

(1) Sudden drop in LH level
(2) Rise in progesterone

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(3) Rise in FSH
(4) Drop in estrogen

39) Acrosome reaction in sperm is triggered by:

(1) Contact of sperm head with zona pellucida
(2) Entry into uterus
(3) Fusion with oolemma
(4) Release of cortical granules

40) Implantation of blastocyst occurs optimally in:

(1) Menstrual phase
(2) Proliferative phase
(3) Secretory (luteal) phase when endometrium is thick and glandular
(4) Ovulatory phase

41) Placenta in humans is formed by fusion of:

(1) Chorionic villi with uterine endometrium
(2) Amnion and chorion only
(3) Yolk sac and allantois
(4) Trophoblast and inner cell mass

42) Gastrulation reorganises the blastocyst into:

(1) Three primary germ layers (ectoderm, mesoderm, endoderm)
(2) Neural tube
(3) Heart primordium
(4) Limb buds

43) The hormone that directly triggers strong uterine contractions for parturition is:

(1) Prolactin
(2) Oxytocin (foetal ejection reflex)
(3) Relaxin
(4) Human placental lactogen

44) blastomeres

(1) Fluid-filled blastocyst
(2) Solid ball of 8–16 blastomeres
(3) Gastrula with three layers
(4) Implanted embryo

45) Number of polar bodies formed during complete oogenesis from one primary oocyte upto

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fertilization:

(1) 1
(2) 2
(3) 4
(4) 0

46) Fertilisation in humans normally occurs in:

(1) Uterus
(2) Ampullary region of fallopian tube
(3) Ovary
(4) Cervix

47) Human placental lactogen (hPL) mainly functions to:

(1) Induce labour
(2) Modify maternal metabolism and prepare for lactation
(3) Prevent polyspermy
(4) Maintain corpus luteum

48) Secondary oocyte remains arrested at metaphase-II until fertilisation to:

(1) Save energy and complete meiosis only if sperm is available
(2) Allow immediate cleavage
(3) Prevent implantation
(4) Form polar bodies early

49) Drop in progesterone level at the end of luteal phase directly causes:

(1) Ovulation
(2) Menstrual flow due to endometrial breakdown
(3) Graafian follicle formation
(4) LH surge

50) If umbilical cord is not clamped after birth, the immediate risk to newborn is:

(1) Continued placental circulation leading to blood loss or infection
(2) No oxygen supply
(3) Immediate lactation
(4) No effect

51) **Statement I:** Spermatogenesis continues throughout life after puberty in males.
Statement II: Oogenesis produces one functional ovum and two polar bodies per primary oocyte.

(1) Only I correct

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(2) Only II correct

(3) Both correct

(4) Both incorrect

52) **Statement I:** Cortical reaction hardens zona pellucida after first sperm entry.
Statement II: This prevents polyspermy in humans.

(1) Only I correct

(2) Only II correct

(3) Both correct

(4) Both incorrect

53) **Statement I:** Placenta acts as an endocrine gland.
Statement II: It secretes hCG, progesterone, estrogen and hPL.

(1) Only I correct

(2) Only II correct

(3) Both correct

(4) Both incorrect

54) **Statement I:** Cleavage divisions increase cell number without increase in embryo size.
Statement II: This helps rapid transport of zygote to uterus before implantation.

(1) Only I correct

(2) Only II correct

(3) Both correct

(4) Both incorrect

55)

Which of the following statements are correct?
(i) Inner cell mass forms the embryo proper.
(ii) Trophoblast forms placenta.
(iii) Amniotic fluid protects embryo from shocks and temperature changes.
(iv) Yolk sac is functional in humans for yolk storage.

(1) (i), (ii) and (iii) only

(2) (ii), (iii) and (iv) only

(3) (i), (iii) and (iv) only

(4) All four

56)

Which statements about menstrual cycle are correct?
(i) LH surge causes ovulation.
(ii) Progesterone is high in luteal phase.
(iii) FSH is high in follicular phase.
(iv) Menstruation occurs due to rise in estrogen.

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(1) (i), (ii) and (iii) only
(2) (i), (ii) and (iv) only
(3) (ii), (iii) and (iv) only
(4) All four

57) **Assertion (A):** Menstruation does not occur during pregnancy.
Reason (R): High levels of progesterone and estrogen suppress GnRH, FSH and LH secretion.

(1) Both A and R true, R explains A
(2) Both true, R does not explain A
(3) A true, R false
(4) A false, R true

58) **Assertion (A):** Placenta is a temporary endocrine gland.
Reason (R): It secretes hormones only in pregnancy.

(1) Both A and R true, R explains A
(2) Both true, R does not explain A
(3) A true, R false
(4) A false, R true

59) Match Column I with Column II:

	Column I (Cell/Hormone)		**Column II (Function)**
A.	Sertoli cells	1.	Secrete testosterone
B.	Leydig cells	2.	Nourish developing spermatids
C.	Trophoblast	3.	Forms chorionic villi of placenta
D.	Corpus luteum	4.	Secretes progesterone in early pregnancy

(1) A-1, B-2, C-3, D-4
(2) A-2, B-1, C-3, D-4
(3) A-3, B-1, C-2, D-4
(4) A-2, B-1, C-4, D-3

60) Match Column I with Column II:

	Column I (Event)		**Column II (Consequence)**
A.	LH surge	1.	Ovulation
B.	hCG secretion	2.	Maintenance of corpus luteum
C.	Oxytocin release	3.	Parturition (labour)
D.	Progesterone withdrawal	4.	Menstruation

(1) A-1, B-2, C-3, D-4

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(2) A-2, B-1, C-3, D-4

(3) A-3, B-1, C-2, D-4

(4) A-2, B-1, C-4, D-3

61) A couple wants biological parenthood where the male has severe oligospermia (very low sperm count). Which ART technique directly addresses this by bypassing natural sperm-egg interaction barriers?

(1) GIFT

(2) ICSI

(3) ZIFT

(4) Standard AI

62) In a high-risk population with multiple partners, which contraceptive simultaneously prevents pregnancy and reduces STD transmission risk most effectively?

(1) Oral pills

(2) Copper-T

(3) Condom

(4) Lactational amenorrhea

63) A woman in her second trimester requests MTP. Why is this riskier than first-trimester MTP under the MTP Act?

(1) Fetus is viable

(2) Procedure becomes more invasive with higher chance of complications

(3) Legal restrictions apply

(4) No medical supervision needed

64) Amniocentesis is legally banned for sex determination but allowed for genetic testing. What high-order ethical-reproductive health conflict does this address?

(1) Population control

(2) Preventing female foeticide while allowing medical necessity

(3) Promoting ART

(4) STD screening

65) A couple with infertility due to blocked fallopian tubes chooses ART. Which technique places the zygote or early embryo directly into the uterus?

(1) GIFT

(2) IUT

(3) ICSI

(4) AI

66) Why is periodic abstinence considered a natural but less reliable method for couples practicing high-order reproductive planning?

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(1) It requires perfect cycle tracking

(2) It is 100 % effective

(3) It involves no cost

(4) It protects against STDs

67) In a rural area with limited medical access, which contraceptive is promoted under RCH programmes for its zero-cost, temporary nature post-delivery?

(1) Copper-T

(2) Lactational amenorrhea

(3) Saheli

(4) Vasectomy

68) A male partner undergoes vasectomy. What physiological change ensures contraception while preserving reproductive hormones?

(1) No sperm ejaculation but semen volume unchanged

(2) Complete loss of libido

(3) Hormonal imbalance

(4) Reversible automatically after 6 months

69) Why do oral contraceptive pills containing estrogen-progesterone prevent pregnancy at multiple levels?

(1) Only inhibit ovulation

(2) Inhibit ovulation + thicken cervical mucus + alter endometrium

(3) Only thicken mucus

(4) Only prevent implantation

70) Which statement best reflects the high-order goal of India's 1951 Family Planning Programme evolving into RCH?

(1) Only reduce population

(2) Achieve total reproductive well-being (physical + emotional + social)

(3) Promote only surgical methods

(4) Ignore STDs

71) Emergency contraceptive pills are most effective when taken within how many hours after unprotected intercourse?

(1) 72 hours

(2) 24 hours

(3) 7 days

(4) 1 month

72) Which contraceptive is least effective against HIV transmission?

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(1) Condom
(2) Oral pills
(3) Copper-T
(4) Both (2) and (3)

73) A couple wants spacing of children after first delivery. Which natural method has good efficacy if followed strictly?

(1) Withdrawal
(2) Periodic abstinence
(3) Lactational amenorrhea (exclusive breastfeeding)
(4) All equal

74) Why is the non-steroidal Saheli preferred over progestogen-only pills in some women?

(1) No effect on menstruation cycle
(2) Weekly dosing improves compliance
(3) It causes permanent infertility
(4) It protects against STDs

75) In a population explosion context, which strategy combines awareness + medical support to achieve small-family norm most effectively?

(1) Only surgical sterilization
(2) RCH programme with education on all contraceptives + postnatal care
(3) Ban on MTP
(4) Promotion of only natural methods

76) What is the primary goal of the Reproductive and Child Health (RCH) program?

(1) Population control
(2) Maternal and child health care
(3) Eradication of STIs
(4) Promotion of contraception only

77) Which method of contraception prevents ovulation?

(1) Condom
(2) Copper-T
(3) Oral contraceptive pills
(4) Diaphragm

78) The surgical method of contraception in males is called:

(1) Tubectomy
(2) Vasectomy

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(3) Hysterectomy
(4) Oophorectomy

79) Which STI is caused by a virus?

(1) Gonorrhea
(2) Syphilis
(3) HIV/AIDS
(4) Chlamydia

80) Amniocentesis is primarily used to detect:

(1) Infertility
(2) Fetal abnormalities diagnosis
(3) STIs
(4) Contraceptive efficacy

81) Which of the following is a natural method of contraception?

(1) IUD
(2) Coitus interruptus
(3) Diaphragm
(4) Implants

82) **Statement I:** Copper-T releases copper ions that suppress sperm motility and prevent fertilization.
Statement II: Copper-T is a barrier method.

(1) Both statement true
(2) Both statement false
(3) Statement I true, statement II false
(4) Statement I false, statement II true

83) **Statement I:** MTP is completely safe in the second trimester.
Statement II: First-trimester MTP requires opinion of only one registered medical practitioner.

(1) Both statement true
(2) Both statement false
(3) Statement I true, II false
(4) Statement I false, II true

84) **Statement I:** ART techniques can help infertile couples have children.
Statement II: All ART methods require surrogate mothers.

(1) Both statement true
(2) Both statement false

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(3) Statement I true, II false

(4) Statement I false, II true

85)

Which of the following statements about STDs are correct?
(i) Most STDs are transmitted through unprotected sexual contact.
(ii) Condoms provide complete protection against all STDs.
(iii) Hepatitis-B and HIV are incurable viral STDs.
(iv) Early detection and treatment can prevent complications like infertility.

(1) (i) & (iv) only

(2) (i), (iii) & (iv)

(3) (ii) & (iii) only

(4) All are correct

86) Which statements are true regarding surgical contraception?
(i) Vasectomy blocks vas deferens.
(ii) Tubectomy blocks fallopian tubes.
(iii) Both are 100 % reversible without surgery.
(iv) Both preserve sexual desire and hormone levels.

(1) (i), (ii) & (iv)

(2) All of these

(3) (i) & (iii) only

(4) (ii) & (iii) only

87)

Which statements correctly describe oral contraceptives?
(i) They inhibit ovulation.
(ii) They are effective immediately after first dose.
(iii) Saheli is a non-steroidal weekly pill.
(iv) They increase risk of STDs.

(1) (i) & (iii) only

(2) (i), (iii) & (iv)

(3) All of these

(4) (i), (ii) & (iii)

88) **Assertion (A):** Condoms are the only contraceptive that also prevents STDs.
Reason (R): They form a physical barrier to both sperm and microbes.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of Assertion (A)

(3) Assertion (A) is true, but Reason (R) is false.

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(4) Assertion (A) is false, but Reason (R) is true.

89) **Assertion (A):** ART has helped many infertile couples.
Reason (R): All infertile cases are due to female factors only.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A)

(3) Assertion (A) is true, but Reason (R) is false.

(4) Assertion (A) is false, but Reason (R) is true.

90) Match Column I (Method) with Column II (Mode of Action):

	Column I		**Column II**
A.	Copper-T	1.	Physical barrier to sperm & pathogens
B.	Oral pills	2.	Inhibits ovulation & thickens mucus
C.	Condom	3.	Inhibits ovulation & thickens mucus
D.	Vasectomy	4.	Blocks sperm transport in vas deferens

(1) A-1, B-2, C-3, D-4

(2) A-2, B-1, C-3, D-4

(3) A-3, B-1, C-2, D-4

(4) A-2, B-3, C-1, D-4

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	2	2	2	2	2	2	1	1	1	1	4	2	2	2	3	2	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	4	4	2	1	1	2	2	1	1	3	3	3	4	3	1	1	2	4
Q.	41	42	43	44	45															
A.	2	2	4	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	1	4	1	4	1	1	2	3	2	1	3	1	4	4	2	1	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	4	2	4	3	3	3	2	3	2	1	2	4	4	1	1	1	4	1
Q.	86	87	88	89	90															
A.	4	3	2	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	4	1	1	3	2	2	1	3	1	3	3	4	3	4	2	4	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	3	2	4	1	2	2	4	3	2	2	2	1	2	2	2	1	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	2	2	2	2	2	1	2	1	3	3	3	3	1	1	1	1	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	2	2	2	1	2	1	2	2	1	4	3	2	2	2	3	2	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	4	3	2	1	1	1	3	4										

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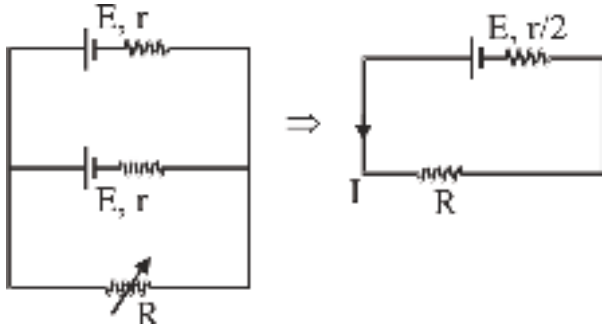
SOLUTIONS

PHYSICS

1) **Explaining :** Find velocity for given time.
Concept : This question is based on average velocity.
Solution :
Part 1: Instantaneous Velocity at t = 3.0 s Velocity (v) is the rate of change of position (x).
$x = A + Bt^2$
$v = \frac{dx}{dt} = \frac{d}{dt}(10 + 3t^2)$
$v = 0 + (2 \times 3)t = \mathbf{6t}$
Now, substitute **t = 3.0 s**:
$v = 6 \times 3.0$
v = 18 m/s
Part 2: Average Velocity between 1.0 s and 3.0 s Average Velocity = (Total Displacement) / (Total Time) $\mathbf{V_{avg} = [x(t_2) - x(t_1)] / [t_2 - t_1]}$
1. Calculate position at **t = 3.0 s** (x_2): $= 10 + 3 \times (3)^2 = 10 + 3 \times 9 = 10 + 27 = \mathbf{37\ m}$
2. Calculate position at **t = 1.0 s** (x_1): $= 10 + 3 \times (1)^2 = 10 + 3 \times 1 = \mathbf{13\ m}$
3. Calculate Average Velocity: $V_{avg} = (37 - 13) / (3.0 - 1.0)$ $V_{avg} = 24 / 2.0$
4. $\mathbf{V_{avg} = 12\ m/s}$
Final Answer : (1)

2) **Question Explanation:**
We are given a circuit in which Two identical cells, each with EMF E – 6V and Internal resistance $r = 2\Omega$ are connected in parallel.
A variable external resistance R is connected across this parallel combination.
We are asked to find the maximum power that can be dissipated across the external resistor R.
Concept Used:
We use the **Maximum Power Transfer Theorem**, which states:
The **maximum power** is delivered to the external load R when the load resistance equals the **internal resistance** of the source (or equivalent internal resistance if multiple sources are present).
When the internal resistance of the source is r, then maximum power occurs when: R = r And the power dissipated is:
$$P_{max} = \frac{E^2}{4r}$$
If there are **two identical cells in parallel**, the **net EMF remains EE**, but the **equivalent internal resistance becomes:.**
$$r_{eq} = \frac{r}{2}$$
Solution :

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Given: E = 6 V, r = 2 Ω
For two identical cells in parallel:

$$r_{eq} = \frac{r}{2} = \frac{2}{2} = 1\Omega$$

By Maximum Power Transfer Theorem:
$R = r_{eq} = 1\Omega$
Now calculate the maximum power:

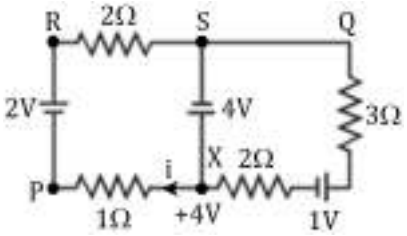
$$P_{max} = \frac{E^2}{4r_{eq}} = \frac{6^2}{4 \times 1} = \frac{36}{4} = 9W$$

Final Answer : (2)

3) **Explaining :** Find the distance of the **second drop** from the **top (the tap)** at that instant.
Concept : This question is based on equation of motion.
Solution :
Step 1: Understand the Timing Total drops involved: 5 Let the time interval between any two drops be **'t'**. When the 5th drop is at the tap (Time = 0), the intervals are: 5th drop: At t = 0
4th drop: Travelled for **t** seconds
3rd drop: Travelled for **2t** seconds
2nd drop: Travelled for **3t** seconds
1st drop: Travelled for **4t** seconds (This one is at the ground)
Step 2: Find the total time for the 1st drop Using the equation: **$H = (1/2) \times g \times T^2$**
$16 = (1/2) \times 10 \times (4t)^2$
$16 = 5 \times 16 \times t^2$
$1 = 5 \times t^2$
$t^2 = 1/5$
Step 3: Calculate the distance of the 2nd drop from the top The 2nd drop has been falling for a time of **3t**.
Distance from top (s) = $(1/2) \times g \times (3t)^2$
$s = (1/2) \times 10 \times 9 \times t^2$
Substitute $t^2 = 1/5$: $s = 5 \times 9 \times (1/5)$
s = 9 meters
Final Result: The second drop is **9 meters** away from the top. The correct option is **(2).**

4) **Explaining :** To find potential difference net points P& Q
Concept : This question is based on KVL
Solution :

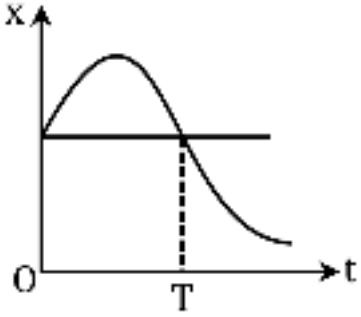
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Let $V_Q = 0V$
Loop RS X PR
$4 - i + 2 - 2i = 0$
$6 = 3i$; $i = 2A$
For 1Ω resistance
$4 - 2(1) = V_P = 2V$
So, $V_P - V_Q = 2V$
Final Answer : (2)

5)

Explaining : Find the graph in which average velocity is zero.
Concept : Where initial & final position of particle is same in time interval 0 to T in that graph displacement is zero as well as average velocity is also zero.
Solution :



Slope of secant is zero so average velocity is also zero.
Final Answer : (2)

6) **Explaining :** We are asked to solve the charge flow through a conductor from t = 2s to t = 8s.

Concept : This question is based on
Area under current v/s time graph represents charge flow.

Solution : Area under the curve represents charge.
$= A_1 + A_2$
$= 4 \times 5 + \frac{1}{2}[5 + 15] \times 2$
$= 40\mu C$

Final Answer : (2)

7) **Explaining :** We are given that the **charge flowing** in a conductor varies with time as:
$Q(t) = at - bt^2$

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We are asked: **Which of the following statements is NOT correct?**

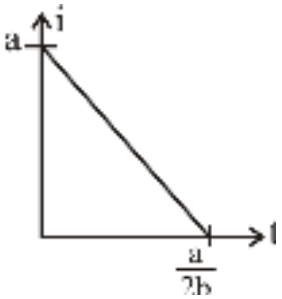
Concept : This question is based on
We use the relation between **current** and **charge**:

$I(t) = \frac{dQ}{dt}$

Differentiate the given expression:

$I(t) = \frac{d}{dt}\left(at - bt^2\right) = a - 2bt$

So, current is a **linear function** of time with **slope = -2b**.



Also:

- At t = 0: I = a
- At $\frac{a}{2b}$: I = 0
- $\frac{dI}{dt} = -2b$: current decreases **linearly**

Solution :

(1) The current decreases linearly with time.

Correct — as I = a − 2bt , it's a straight line with negative slope.

(2) Current increases linearly with time.

Incorrect — current is **decreasing**, not increasing. This is the **false** (NOT correct) statement.

(3) Current falls to zero after time $t = \frac{a}{2b}$

Correct — from a − 2bt = 0 ⇒ t = $\frac{a}{2b}$

(4) Current changes at a rate -2b.

Correct — $\frac{dI}{dt} = -2b$

Final Answer : (2)

8) **Explain Question:**

Find which bulb glow with maximum brightness.

Concept:

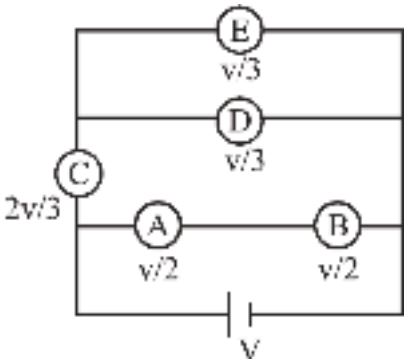
Brightness of bulb $(P) = \frac{V^2}{R}$.

Solution:

In given question all bulbs are identical i.e. having same resistance.

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So $P \times V^2$



Potential difference of (C) is
Maximum hence bulb (C) lights with maximum brightness.

Final Answer: Option (2)

9) **Explaining :** Calculate displacement in given time.
Concept : This question is based on apply equation of motion.
Solution : From $S = ut + \frac{1}{2}at^2$

$S_1 = \frac{1}{2}a(p-1)^2$ and $S_2 = \frac{1}{2}ap^2$

[As u = 0]

From $S_n = u + \frac{a}{2}(2n-1)$

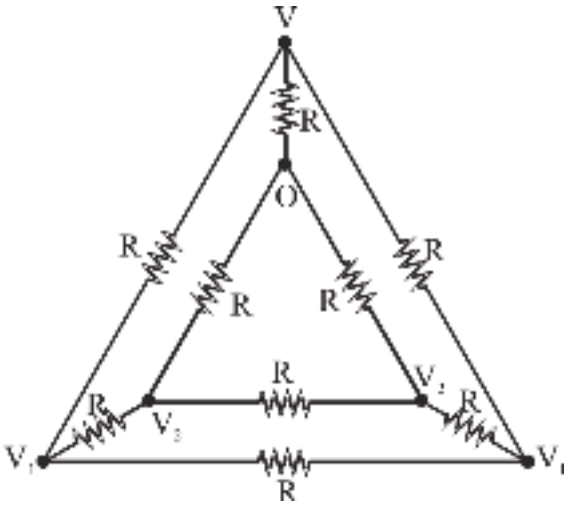
$S_{(p^2-p+1)^{th}} = \frac{a}{2}[2(p^2-p+1)-1]$

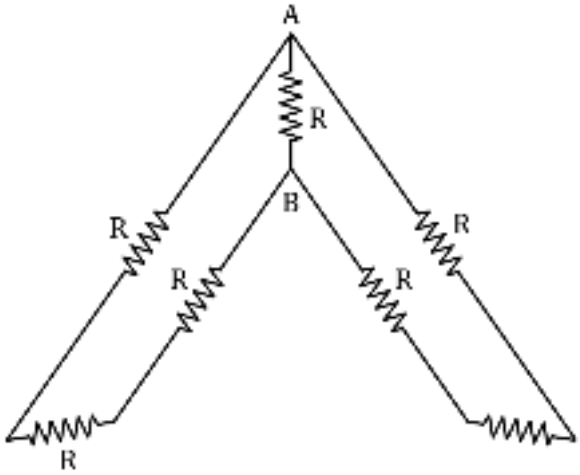
$S_{(p^2-p+1)^{th}} = \frac{a}{2}[2p^2-2p+1]$

It is clear that $S_{(p^2-p+1)^{th}} = S_1 + S_2$

Final Answer : (1)

10) **Explaining :** To find equivalent resistance between A and B.
Concept : Equivalent resistance
Solution :





$R_{eq} = \frac{3R}{5}$

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Final Answer : (1)

11) **Explaining :** We have to find stopping distance & time.
Concept : This question is based on stopping distance (equation of motion).
Solution :
Step 1: Given Data & Unit Conversion
Initial Velocity (u) = 72 km/h
Final Velocity (v) = 0 m/s (Since the car stops)
Acceleration (a) = **–5 m/s²** (Negative sign because it is retardation)
Step 2: Calculate Stopping Time (t) Using the first equation of motion: **v = u + at**
0 = 20 + (-5) × t
5t = 20 ⇒ **t = 4 seconds**
Step 3: Calculate Stopping Distance (s) Using the third equation of motion:
$v^2 = u^2 + 2as$
$(0)^2 = (20)^2 + 2 \times (-5) \times s$
s = 40 meters
Final Result: Time taken is **4 seconds** and Stopping Distance is **40 meters**. The correct option is **(1)**.

12) **QUESTION EXPLANATION:**
You are given a **graph between voltage (V)** on the **y-axis** and **current (i)** on the **x-axis** for a circuit containing:

A. A **battery of EMF** ε and **internal resistance** r
B. Connected to a variable external resistor R

The graph is:

A. A **straight line** (implying linear relationship)
B. Has a **negative slope**
C. Y-intercept (voltage when current is zero) = **10 V**
D. X-intercept (current when voltage is zero) = **2 A**

You're to determine the **correct option** among:

A. Internal resistance of battery is 5 Ω
B. EMF of the battery is 20 V
C. Maximum current which can be taken from the battery is 4 A
D. V–i graph can never be a straight line as shown in figure

CONCEPT:
The voltage across the external resistance is:
$V = \varepsilon - ir$
This is a **linear equation** in the form:
$V = -r(i) + \varepsilon$

A. Y-intercept = ε
B. Slope = −r
C. X-intercept $i = \frac{\varepsilon}{r}$ (maximum current when external resistance is 0)

SOLUTION: From the graph:

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A. **Y-intercept** = 10 V → ε =10 V
B. **X-intercept** = 2 A → ε = i_{max}(r)

$$10 = 2r \Rightarrow r = 5\ \Omega$$

Now go through the options:
Option (1): Internal resistance is 5 Ω **Correct**
Option (2): EMF is 20 V **Incorrect**, it's 10 V
Option (3): Maximum current is 4 A **Incorrect**, it's 2 A
Option (4): V–i graph can never be a straight line **Incorrect**, it is a straight line, as shown by V – ε − ir **FINAL ANSWER:**
(1) Internal resistance of battery is 5 Ω

13)

Explaining : Assertion–Reason Question, we have to select correct alternative.
Concept : Positive and negative sign with velocity represents direction of motion.
Solution : Either velocity is positive or negative distance travelled by a body is always positive, Hence assertion is wrong.
Final Answer : (4)

14) **Explaining :** Find magnitude of relative velocity
Concept : This question is based on slope of x- t graph gives velocity & $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$
Solution :

$$v_A = -\frac{8}{2} = -4 m/s; v_B = \frac{6}{6} = 1 m/s$$

$$v_{BA} = [1 - (-4)] = 5 m/s$$

Final Answer : (2)

15) **Explaining :** We have to find distance at what distance from the **top of the tower** will the two particles collide.
Concept : This question is based on motion under gravity.
Solution :
Step 1: Given Data
Height of Tower (H) = 100 m
Initial velocity of Particle A (u) = 50 m/s (Upwards)
Initial velocity of Particle B = 0 m/s (Dropped)
Acceleration due to gravity (g) = 10 m/s^2
Step 2: Find the time of meeting (t) Using the concept of Relative Motion:
Relative Velocity (V_{rel}) = Velocity of A + Velocity of B
V_{rel} = 50 + 0 = 50 m/s
Relative Distance = 100 m
Time (t) = Distance / Relative Velocity
t = 100 / 50 = **2 seconds**
Step 3: Find the distance fallen by Particle B from the top Particle B is dropped (initial speed = 0), so we use the second equation of motion:
Distance (s) = (1/2) × g × t^2
s = (1/2) × 10 × $(2)^2$
s = 5 × 4
s = 20 meters
Conclusion: The particles will meet at a distance of **20 meters from the top**.

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Final Answer : (2)

16) **Explaining :** To find the resistance of voltmeter in the circuit shown.
Concept : This question is based on circuit analysis.
Solution :
$V_{24} = 30 - 6 = 24V$

$i = \frac{V_{24}}{R_{24}} = \frac{24V}{24\Omega} = 1A$

$V_V = 6V = i(R) = (1)\left(\frac{12R_V}{12 + R_V}\right)$

$2R_V = 12 + R_V \Rightarrow R_V = 12\Omega$

Final Answer : (2)

17) **Explaining :** Find the acceleration (a) of the particle in terms of its velocity (v).
Concept : This question is based on calculus.
Solution :
$t = 2x^2 + 3x$

$\frac{d}{dt}(2x^2 + 3x)\ 1 = (4x + 3) \times \left(\frac{dx}{dt}\right)$

We know that velocity (v) = $\frac{dx}{dt}$.

$1 = (4x + 3) \times v$

$\therefore v = \frac{1}{(4x + 3)}$ ....(Equation 1) OR

$(4x + 3) = \frac{1}{v}$ ....(Equation 2)

Acceleration (a) = $\frac{dv}{dt}$. Using Equation 1: $v = (4x + 3)^{-1}$

$a = \frac{dv}{dt} = \frac{d}{dt}[(4x + 3)^{-1}]$

$a = -1 \times (4x + 3)^{-2} \times \left(4 \times \frac{dx}{dt}\right)$

$(4x + 3)^{-1} = v$, so $(4x + 3)^{-2} = v^2$.

$\frac{dx}{dt} = v$.

$a = -1 \times (v^2) \times (4 \times v)$ **$a = -4v^3$**

Final Answer : (3)

18) **Explaining :** To find net resistance increases or decreases.
Concept : Temperature dependence of resistivity.
Solution :

$R_1' = R_1(1 + \alpha T)$ $\quad$ $R_2' = R_2(1 - \beta T)$

$\Delta R_1' = \alpha R_1 T$ $\quad$ $\Delta R_2' = -\beta R_2 T$

$R_{eq} = R_1' + R_2'$

$\Delta R_{eq} = \Delta R_1' + \Delta R_2'$

$= T(\alpha R_1 - \beta R_2)$

If $\alpha R_1 > \beta R_2$

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$\Delta R_{eq} \rightarrow +ve$
So, net resistance increases with increase in temperature.
Final Answer : (2)

19) **Explaining :** To find v-t graph for given a - t graph
Concept : This question is based on Key point :
(i) If a is constant then v - t graph is line as
(ii) change in velocity = area under a-t graphs.
Solution : Area of a – t graph gives change in velocity
u = 0
area = 10 (for t = 0 to 2 sec)
So, velocity increases from 0 to 10 in 2 sec, linearly.
Final Answer : (2)

20) **Explaining :** Find time taken by coin to reach at floor.
Concept : This question is based on relative motion.
Solution :
$t = \sqrt{\frac{2S_{rel}}{a_{rel}}} = \sqrt{\frac{2(2)}{10-9}} = 2$ sec
Final Answer : (2)

21) **Explaining :** Select incorrect statement
Concept : Application of calculus in general motion.
Solution :
(1) slope of x-t graph is velocity if velocity is constant then acceleration is zero.
(2) At top most point of vertical motion under gravity v = 0 but a = g
(3) For constant velocity, | average veloriciy | = average speed
(4) If velocity & acceleration both have opposite sign the speed reduces.
Final Answer : (4)

22) **QUESTION EXPLANATION:**
We are given a **straight conductor of uniform cross-section** carrying a **steady current** I.
We are asked to find the **total momentum** of **all free electrons** in the conductor.
We are also told that:

A. $S = \frac{e}{m}$ is the **specific charge** of the electron (i.e., charge-to-mass ratio).

CONCEPT:
Drift velocity v_d is the average velocity with which free electrons move under an electric field in a conductor carrying current I.
Even though the electrons move randomly due to thermal motion, there's a **net drift** in one direction.
The **total momentum** of all free electrons is:
Total momentum = Total mass of all free electrons × Drift velocity
We use:

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A. Drift velocity: $v_d = \frac{I}{nAe}$

B. Number of free electrons in volume V:

N = nV = nAℓ (for a conductor of length ℓ and area A)

A. Total mass of electrons: M = N • m = nAℓ • m

So,

Total momentum = $Mv_d = (nA\ell)(m)\left(\frac{I}{nAe}\right)$ **SOLUTION:**

Let's compute step-by-step:

Total momentum = $(nA\ell)(m)\left(\frac{I}{nAe}\right)$

Total momentum = $(m\ell)\left(\frac{I}{e}\right)$

Rewriting using **specific charge** $S = \frac{e}{m} \Rightarrow \frac{1}{S} = \frac{m}{e}$:

Total momentum = $= \ell\left(\frac{m}{e}\right) I = \left(\frac{I\ell}{S}\right)$ **FINAL ANSWER :**

Total momentum of all free electrons = $\frac{I\ell}{s}$

23) **Explaining :** Find height of balloon from ground.

Concept : This question is based on When particle is dropped from moving object then its initial velocity is equal to velocity of moving object at the instant of drop.

Solution :

$S = 10(7) - \frac{1}{2}(10)(7)^2$

$S = 70 - 5(49)$

$S = -175$

h = 175m

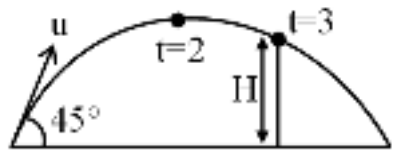
Final Answer : (4)

24) **Explaining :** Find value of H for given situation.

Concept : This question is based on ground–ground projectille motion.

Solution :

$T = \frac{2u_y}{g} = 4$



$\Rightarrow u_y = \frac{40}{2} = 20$ m/s

$y = u_y\Delta t - \frac{1}{2}g(\Delta t)^2$

$\Rightarrow H = 20 \times 3 - 5 \times 9$

$= 60 - 45$

$= 15$ m

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Final Answer : (4)

25) **Explaining :** Calculate time when $\vec{v}$ and $\vec{a}$ are perpendicular to each other.
Concept : Dot product of two perpendicular vectors is zero.
Solution : $\vec{r} = (t^2 - 4t + 6)\hat{i} + t^2\hat{j}$;

$$\vec{v} = \frac{d\vec{r}}{dt} = (2t-4)\hat{i} + 2t\hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 2\hat{j}$$

If $\vec{a}$ and $\vec{v}$ are perpendicular, $\vec{a}.\vec{v} = 0$
$(2\hat{i} + 2\hat{j}).((2t-4)\hat{i} + 2t\hat{j}) = 0$
8t - 8 = 0, t = 1 sec.
Final Answer : (2)

26) **QUESTION EXPLANATION:**

A. An **electric kettle** has **two heating coils**.
B. Time taken to boil water:
   A. **Coil A alone**: 5 minutes
   B. **Coil B alone**: 10 minutes
C. When **both coils are connected in parallel**, we are asked to find the **new time** to boil the same amount of water.

 Heat required to boil water is constant.
Let total energy required to boil water be Q.
From the formula:

$$Q = Pt \Rightarrow \text{Time} = \frac{Q}{P}$$

If coil A alone boils in 5 minutes, then:

A. Power of coil A: $P_A = \frac{Q}{5}$

If coil B alone boils in 10 minutes:

A. Power of coil B : $P_B = \frac{Q}{10}$

When both coils are connected in parallel:

A. Total power: $P = P_A + P_B = \frac{Q}{5} + \frac{Q}{10} = \frac{3Q}{10}$

So total time required:

$$t = \frac{Q}{P} = \frac{Q}{\frac{3Q}{10}} = \frac{10}{3} \text{ minutes } = 3 \text{ minute and } 20 \text{ seconds}$$

FINAL ANSWER:
(1) 3 min 20 sec
So when both coils are connected in parallel, **water boils in 3 minutes and 20 seconds**.

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27) **QUESTION EXPLANATION:**
You are told:

A. In a **meter bridge**, the **null point is at 25 cm from A** when balancing unknown resistance R against standard resistance S.
B. Then, a **10 Ω resistor is connected in parallel with S**.
C. After this, the **null point shifts to the midpoint**, i.e., **50 cm from A**.
D. You are to find the value of R.

CONCEPT:
The meter bridge is based on the **Wheatstone bridge principle**:
$$\frac{R}{S} = \frac{\ell}{100-\ell}$$
Where:

A. ℓ is the distance of the null point from end A
B. R, S are resistances in the two arms of the bridge

SOLUTION:
1. Initial condition (null point at 25 cm):
$$\frac{R}{S} = \frac{25}{75} = \frac{1}{3} \quad (\text{Equation 1}) \Rightarrow R = \frac{S}{3}$$
2. Now a 10 Ω resistor is connected in parallel with S:
Let the **effective resistance** S′ be:
$$\frac{1}{S'} = \frac{1}{S} + \frac{1}{10} \Rightarrow S' = \frac{10S}{S+10}$$
Now, the null point shifts to **50 cm**, so:
$$\frac{R}{S'} = \frac{50}{50} = 1 \quad (\text{Equation 2}) \Rightarrow R = S'$$
Now substitute $R = \frac{S}{3}$ from Equation 1 into Equation 2:
$$\frac{S}{3} = \frac{10S}{S+10}$$
Multiply both sides by 3(S+10):
$S(S + 10) = 30S \Rightarrow S^2 + 10S = 30S \Rightarrow S^2 - 20S - 0 \Rightarrow S(S - 20) = 0$
So, S = 0 or S = 20. Discarding S = 0, we get:
$$S - 20\ \Omega \Rightarrow R - \frac{S}{3} = \frac{20}{3} \approx 6.67\ \Omega$$ **FINAL ANSWER:**
$$\boxed{R = \frac{20}{3}\ \Omega = 6.67\ \Omega}$$

28) **Explaining :** Calculate height h

Concept : This question is based on time of flight $T = t_A + t_D = \frac{2u}{g}$
& apply $S = ut + \frac{1}{2}at^2$
Solution :
$$t_1 + t_2 = \frac{2u}{g}$$
$$u = \frac{g(t_1 + t_2)}{2}$$
$$S_y = U_y t + \frac{1}{2}a_y t^2$$

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$$h = \frac{g(t_1 + t_2)}{2} t_1 - \frac{1}{2} g t_1^2$$

$$h = \frac{g t_1^2}{2} + \frac{g t_1 t_2}{2} - \frac{1}{2} g t_1^2$$

$$h = \frac{1}{2} g t_1 t_2 = \frac{1}{2} \times 10 (2) (8) = 80 \text{ m}$$

Final Answer : (2)

29) **Explaining :** To find the ratio of drift velocity

Concept : This question is based on drift velocity

Solution :

Since I → Same ; $v_d \propto \frac{1}{A}$

Also $R_1 = R_2 \Rightarrow \frac{\rho L}{A_1} = \frac{\rho (2L)}{A_2} \Rightarrow \frac{A_1}{A_2} = \frac{1}{2}$

$\Rightarrow \frac{(v_d)_1}{(v_d)_2} = \frac{A_2}{A_1} = \frac{2}{1}$

Final Answer : (2)

30) **Explaining :** Match the column

Concept : This question is based on ground–ground projectile motion.

$$T = \frac{2U_y}{g}$$

$$H = \frac{U_y^2}{2g}$$

$$R = \frac{2U_x U_y}{g}$$

Solution :

$\vec{U} = U_x \hat{i} + U_y \hat{j}$

$\vec{U} = 30\hat{i} + 40\hat{j}$

$$T = \frac{2U_y}{g} = \frac{2 \times 40}{10} = 8 \text{ sec.}$$

$$H = \frac{U_y^2}{2g} = \frac{40 \times 40}{20} = 80 \text{ m}$$

$$R = \frac{2U_x U_y}{g} = \frac{2(30)(40)}{10} = 240 \text{ m}$$

& Horizontal velocity remains constant.

Final Answer : (1)

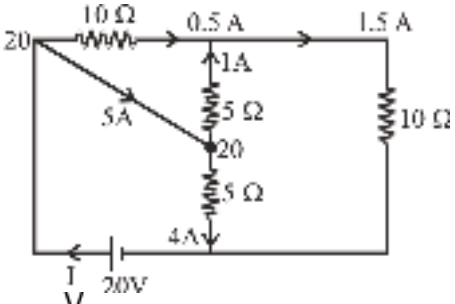
31) **Explaining :** To find the current in branch AB of circuit.

Concept : This question is based on Network analysis.

Solution : $R_{eq} = 5 \,||\, (10 + 10/3)$

$R_{eq} = 40/11\ \Omega$

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$$i = \frac{V}{R_{eq}} = 5.5A$$

Final Answer : (1)

32) **Explaining :** Calculate time of collision.
Concept : This question is based on when relative acceleration is zero then for the collision of two particles the direction of relative velocity is along the line joining of particles.
Solution :
along line joining
$V_{BA} = 30$ m/s

$$t = \frac{210}{30} = 7 \text{ sec}$$

Final Answer : (3)

33) **Explaining :** To find resistivity of wire **Concept : This question is based on** resistance and resistivity
Solution :

$$R = \frac{V}{I} = \tan 53^{o} = \frac{4}{3}\Omega \; ; r = 10^{-3} \text{ m}$$

$$\frac{4}{3} = \frac{\rho \ell}{A} \Rightarrow \rho = \frac{4}{3}\frac{A}{\ell} = \frac{4}{3}\frac{\left(3.14 \times 10^{-6}\right)}{(0.5)}$$

$\rho \approx 8 \times 10^{-6} \Omega^{m}$
Final Answer : (3)

34) **Explaining :** To determine the value of R_1 and R_2. **Concept :** This question is based on cell.
Solution :
When $R_3 = 0$ When $R_3 = \infty$
$E = I(R_1 + R_2)$ $E = I(R_1 + 2R_2)$
$E = 9(R_1 + R_2)$ $E = 6(R_1 + 2R_2)$
$9(R_1 + R_2) = 6(R_1 + 2R_2)$
$3R_1 = 3R_2 \Rightarrow R_1 = R_2$
Final Answer : (3)

35) **Explaining :** To solve for resistance R in the circuit shown.
Concept : This question is based on $P = I^2R$, So, $I = \sqrt{\frac{P}{R}}$
Solution :

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Current in 6Ω $I = \sqrt{\frac{P}{R}} = \sqrt{\frac{6}{6}} = 1A$

$I = \frac{12}{6 + \frac{8R}{8+R}}$ $1 = \frac{12}{48 + 14R}(8 + R)$

R = 24Ω

Final Answer : (4)

36) **Explaining :** To choose the correct circuit

Concept : This question is based on experimental skills

Solution :

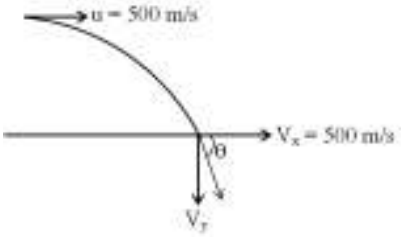
The circuit represents correct connection of R_1 (high resistance) in series and R_2 (small resistance) as shunt for measurement of V & I respectively as V & A

Final Answer : (3)

37) **Explain the Question :** Asking about the angle from horizontal at which the bomb stripes the ground.

Concept : This question is based on horizontal projectile.

Solution :



V_x = 500 m/s

$V_y = u_y + at$

$= 0 + 10 \times 10$

= 100 m/s

$\tan\theta = \frac{100}{500} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{5}\right)$

Final Answer : 1

38) **Explaining :** To solve the value of x for which bridge is balanced. **Concept :** This question is based on for balanced WSB, resistance ratios are equal.

Solution :

For balanced bridge

$\frac{20x}{20 + x} = 16$

$320 + 16x = 20x$

$x = 80\ \Omega$

Final Answer : (1)

39) **QUESTION :**

Given a **galvanometer** with:

A. Resistance $R_g = 100\,\Omega$

B. Full-scale deflection current $I_g = 50\,\mu A = 50 \times 10^{-6}\,A$

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You're asked to evaluate four statements:
(A) Resistance needed to use it as a **voltmeter of range 50 V** is $10^6\,\Omega$
(B) Same as above but says 10^5
(C) Resistance needed to use it as an **ammeter of range 10 mA** is $0.5\,\Omega$
(D) Same as above but says $1.0\,\Omega$ **CONCEPTS:**
 Voltmeter Conversion:
To convert galvanometer to a voltmeter of range V = 50 V, use a **series resistance** R

$$R = \frac{V}{I_g} - R_g$$

 Ammeter Conversion:
To convert it to an ammeter of range I = 10 mA, use a **shunt resistance R_s** in parallel:

$$R_s = \frac{I_g R_g}{I - I_g}$$

CALCULATIONS:
 For Voltmeter:

$$R = \frac{50}{50 \times 10^{-6}} - 100 = 1,000,000 - 100 = 999,900\ \Omega \approx \boxed{10^6\ \Omega}$$

So **(A) is correct**, **(B) is wrong** (off by a factor of 10) **For Ammeter**

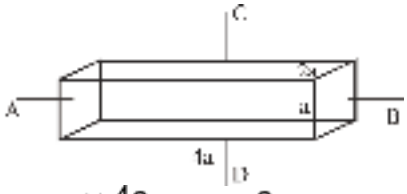
$$I = 10\,mA = 0.01\,A \quad \Rightarrow \quad R_s = \frac{(50 \times 10^{-6}) \cdot 100}{0.01 - 50 \times 10^{-6}} = \frac{0.005}{0.00995} \approx \boxed{0.5025\,\Omega}$$

So **(C) is correct**, **(D) is wrong** **FINAL ANSWER:**

A. **(A)** Correct
B. **(B)** Wrong
C. **(C)** Correct
D. **(D)** Wrong

So the correct option is:
(2) Only A, C

40) **Explaining :** The question is asking about the resistance of conductor across AB and CD.
Concept : This question is based on $R = \frac{\rho \ell}{A}$
Solution :

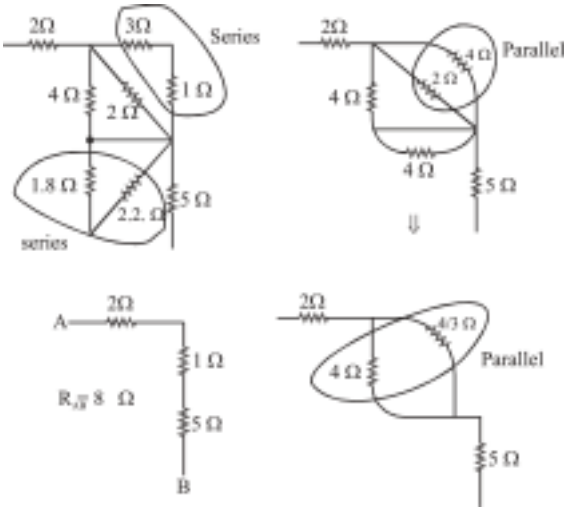


$$x = \frac{\rho \times 4a}{2a^2},\ y = \frac{\rho a}{8a^2}$$

So, $\frac{x}{y} = \frac{2\rho/a}{\rho/8a} = 16$
Final Answer : (4)

41) **Explaining :** To solve the equivalent resistance of the network between A and B. **Concept : This question is based on** The resistance which are open i.e. there one end is not connected to main circuit, they are not the part of circuit.
Solution :

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Final Answer : (2)

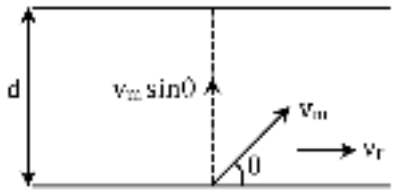
42) **Explaining :** We have to calculate time for swimmer in given situation. **Concept :** This question is based on River-man Problem.

Solution :

(a) Width of River (d) = 500 m = 0.5 km

Velocity of River (Vr) = 5 km/hr

Velocity of swimmer wrt. Water (V_m) = 3 km/hr



$$\text{time} = \frac{\text{width}}{\text{velocity component perpendicular to flow}}$$

$$t = \frac{0.5}{3\sin\theta} \text{ hrs.}$$

(b) $t_{min} = \frac{0.5}{3 \times \sin 90^\circ} = \frac{0.5}{3} \text{hr}$

$t_{min} = \frac{1}{6} \times 60$

= 10 min.

Final Answer : (2)

43) **QUESTION EXPLANATION:**

We are given:

A. **n identical cells** in series.
B. Each cell has:
    A. EMF = E
    B. Internal resistance = r
C. Two cells (A and B) are connected **with reversed polarity**, so they **oppose** the rest.
D. Total number of cells n > 4
E. We need to find the **potential difference across either reversed cell (A or B).**

 CONCEPT:

 Net EMF in the loop:

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A. Out of n cells:
   A. n − 2 cells are aligned properly → each contributes +E
   B. 2 reversed cells contribute −E each
B. So net EMF:

$\varepsilon_{net} = (n-2)E - 2E = (n-4)E$

Total internal resistance :

All n cells are still in series:

$R_{total} = nr$

Current in the loop:

Using Ohm's law:

$$I = \frac{\varepsilon_{net}}{R_{total}} = \frac{(n-4)E}{nr}$$

SOLUTION: We now want to find the **terminal potential difference across a reversed cell** (say cell A).

Here's the key idea:

A. In a **reversed cell**, the EMF opposes the current.
B. But **current still flows from the negative terminal to the positive terminal** of that cell (because it's reversed).
C. So, the **potential difference across the terminals** becomes:

$V = E + Ir$

Now, plug in the current:

$$V = E + \left(\frac{(n-4)E}{nr}\right)r = E + \frac{(n-4)E}{n}$$

Simplify :

$$V = E\left(1 + \frac{n-4}{n}\right) = E\left(\frac{n+(n-4)}{n}\right) = E\left(\frac{2n-4}{n}\right)$$

FINAL ANSWER: (4)

$$\boxed{V = \frac{(2n-4)E}{n} = 2E\left(1 - \frac{2}{n}\right)}$$

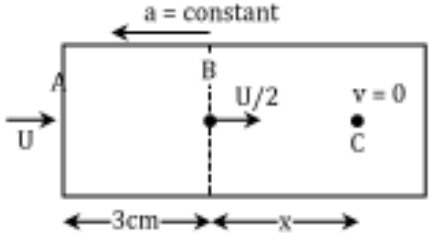
This is the **terminal potential difference across either reversed cell A or B**, taking the correct direction of current and polarity into account.

44) **Explaining :** Calculate distance travelled by bullet.

Concept : This question is based on equation of motion.

Solution :

- Apply 3rd equation of motion.



For AB

$v^2 = U^2 + 2as$

$$\left(\frac{U}{2}\right)^2 = U^2 - 2a(3)$$

$$\frac{U^2}{4} = U^2 - 6a$$

$$6a = U^2 - \frac{U^2}{4}$$

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$6a = \frac{3U^2}{4} \Rightarrow a = \frac{U^2}{8}$

For BC

$v^2 = U^2 + 2as$

$0 = \left(\frac{U}{2}\right)^2 - \frac{2 \times U^2}{8}x$

$\frac{U^2}{4}x = \frac{U^2}{4}$

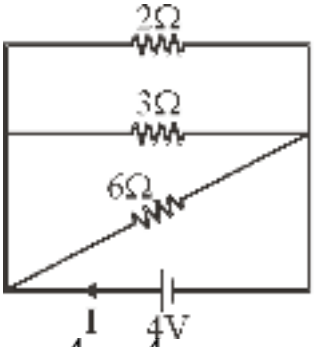
x = 1 cm.

Final Answer : (2)

45) **Explaining :** The question is asking for current through battery for the circuit shown.

Concept : 5Ω and 4Ω get short circuited.

Solution :



$R_{eq} = 1\ \Omega$

$I = \frac{4}{R_{eq}} = \frac{4}{1} = 4A$

Final Answer : (4)

CHEMISTRY

46)

Explaining : - Find reaction between k_1 k_2 & k_3

Concept : This question is based on rate of reaction.

Solution : Rate of reaction

$= -\frac{1}{2}\frac{d[A]}{dt} = \frac{d[B]}{dt} = \frac{1}{3}\frac{d[C]}{dt}$

or $\frac{k_1[A]^2}{2} = \frac{k_2[A]^2}{1} = \frac{k_3[A]^2}{3}$

$\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

Final Answer : (4)

47)

Explaining : - Find unit of k of nth order kinetics.

Concept : This question is based on unit of rate constant.

Solution : Rate of reaction = $K[conc.]^n$

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$$K = \frac{\text{Rate}}{[\text{conc.}]^n} = \frac{Ms^{-1}}{M^n} = \frac{\text{mol}\,L^{-1}s^{-1}}{\text{mol}^n L^{-n}} = \text{mol}^{1-n}\,L^{n-1}\,s^{-1}$$

Final Answer : (4)

48)

Explaining : - Find dependency of rate constant.
Concept : This question is based on rate constant.
Solution : Arrhenius eq^n. rate constant $K = Ae^{-Ea/RT}$
Final Answer : (1)

49)

Explaining : - Find molecularity of a complex reaction
Concept : This question is based on complex reaction molecularity.
Solution : Molecularity of a complex reaction is meaningless.
Final Answer : (4)

50)

Explaining : - Find order with A_2 and B_2
Concept : This question is based on rate law expressions.
Solution :
Rate of reaction = $K[A_2]^a[B_2]^b$
$0.04 = K(0.2)^a(0.2)^b$ ...(i)
$0.04 = K(0.1)^a(0.4)^b$ ...(ii)
$0.08 = K(0.2)^a(0.4)^b$ ...(iii)

$\frac{(i)}{(iii)}$ $\frac{1}{2} = \left(\frac{1}{2}\right)^b$; b = 1

$\frac{(i)}{(ii)}$ $1 = 2^a \times \left(\frac{1}{2}\right)^b$; a = 1

Final Answer : (1)

51)

Explaining : - Find rate of formation of B.
Concept : This question is based on rate law expressions.
Solution : Rate of reaction = $k[A]^2$

Rate of disappearance of A = $-\frac{d[A]}{dt}$

Rate of appearance of B = $+\frac{d[B]}{dt}$

Rate of reaction = $-\frac{d[A]}{dt} = +\frac{d[B]}{dt} = k[A]^2$

Rate of appearance of B after double the conc. of A.

$\frac{d[B]}{dt} = k[2A]^2$

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Final Answer : (4)

52)

Explaining : - Find half-life
Concept : This question is based on half-life of nth order.
Solution : $$t_{1/2} = \frac{1}{(n-1)k}\left[\frac{2^{n-1}-1}{a^{n-1}}\right]$$
Final Answer : (1)

53)

Explaining : - Find order of reaction.
Concept : This question is based on unit of reaction.
Solution : Rate = $k[conc.]^0 = k$; for zero order reaction.
Final Answer : (1)

54)

Explaining : - Find type of reaction
Concept : This question is based on Pseudo first order reaction.
Solution : Conc. of water (solvent) remains constant so rate is independent from conc. of H_2O.
Final Answer : (2)

55)

Explaining : Find rate of reaction
Concept : This question is based on rate of reaction.
Solution : $r_1 = k[A]^2[B]$; $r_2 = k[2A]^2[2B] = 8\ r_1$
Final Answer : (3)

56)

Explaining : - Find relation between $t_{87.5\%}$ & $t_{75\%}$
Concept : This question is based on half-life of first order reaction.
Solution :
$t_{75\%} = 2 \times t_{50\%}$
$100 = 2 \times t_{50\%}$
$t_{50\%} = 50$ minutes
Now $t_{87.5\%} = 3 \times t_{50\%}$
$= 3 \times 50$
$= 150$ min
Final Answer : (2)

57)

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Explaining : Find rate constant of first order reaction.
Concept : This question is based on rate constant.
Solution : $K = \frac{0.693}{t_{1/2}} = \frac{0.693}{480} = 1.44 \times 10^{-3} s^{-1}$
Final Answer : (1)

58)

Explaining : Find graph by given reaction.
Concept : This question is based on activation energy
Solution :
First step is slow (require more activation energy) second step is fast (less activation energy) and overall reaction exothermic, so product energy level should be less as compared to reactants.
Final Answer : (3)

59)

Explaining : If the value of equilibrium constant for a particular reaction is 1.6×10^{12}, then at equilibrium, the system will contain
Concept : This question is based on Application of equilibrium constant.
Solution : Higher the K, Higher the concentration of product at equilibrium.
Final Answer : (1)

60)

Explaining : If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K,
the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be
Concept : This question is based on application of equilibrium constant.
Solution : Multiply given eq. with $\frac{1}{2}$. So new K becomes $K^{1/2}$.
Final Answer : (4)

61)

Explaining : In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be
Concept : This question is based on application of equilibrium constant.
Solution : Value of K_{eq} depends only on Temperature.
Final Answer : (4)

62)

Explaining : The equilibrium constant for the reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 2000 K. In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant, in presence of the catalyst, at 2000 K is:
Concept : This question is based on application of equilibrium constant.

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Solution : K_{eq} is independent of catalyst.
Final Answer : (2)

63)

Explaining : Partial pressure of O_2 in the reaction $2Ag_2O$ (s) ⇌ 4 Ag (s) + O_2 (g) is
Concept : This question is based on Law of mass action.
Solution : $K_p = P_{O_2}$
Final Answer : (1)

64)

Explaining : A quantity of PCl_5 was heated in a 10 dm^3 vessel at 250°C : PCl_5 (g) ⇌ PCl_3 (g) + Cl_2(g). At equilibrium, the vessel contains 0.1 mole of PCl_5 and 0.2 mole of Cl_2. The equilibrium constant of the reaction is.
Concept : This question is based on Law of mass action.
Solution :

PCl_5 ⇋ PCl_3 + Cl_2
0.1 mol 0.2 0.2

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} \Rightarrow K_c = \frac{\frac{0.2}{10} \times \frac{0.2}{10}}{\frac{0.1}{10}} = 0.04$$

Final Answer : (1)

65)

Explaining : A + B ⇌ C + D. If initially the concentrations of A and B are equal but at equilibrium, concentration of D will be twice that of A, then what will be the equilibrium constant of the reaction ?
Concept : This question is based on Law of mass action.
Solution :

	A	+	B	⇌	C	+	D
t_0	a		a		–		–
t_{eq}	a–x		a–x		x		x

[D] = 2 × [A] ⇒ x = 2 × (a – x)
x = 2a/3

$$K_c = \frac{[C][D]}{[A][B]} = \frac{\frac{2a}{3} \cdot \frac{2a}{3}}{\frac{a}{3} \cdot \frac{a}{3}}$$

K_c = 4
Final Answer : (4)

66) **Explaining :** True statement among the following.
Concept : This question is based on active mass.
Solution :
(i) Active mass of pure liquids and solids are taken unity.
(ii) Active mass of gases can be expressed is terms of atm.

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(iii) Law of mass action is applicable on reversible reaction
Final Answer : (4)

67) **Explaining :** The equilibrium constant (K_c) for the reaction $HA + B \rightleftharpoons BH^+ + A^-$ is 100. If the rate constant for the forward reaction is 10^5, then rate constant for the backward reaction is
Concept : This question is based on equilibrium constant (K_c)
Solution :

$$K_c = \frac{K_f}{K_b}$$

$$100 = \frac{10^5}{K_b}$$

$K_b = 10^3$
Final Answer : (2)

68) **Explaining :** True statement.
Concept : This question is based on Type of reaction.
Solution :
(1) In exothermic reaction, heat is evolved during reaction.
(2) Product is more stable then reactant.
Final Answer : (4)

69)

Explaining : comparision of pH of solution.
Concept : This question is based on hydrolysis of salt.
Solution :

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

Lower the pK_a, better the acid $HClO_4$ has highest acidic strength.
Final Answer : (2)

70)

Explaining : pK_a of weak acid (HA) and pK_b of weak base (BOH) are 3.2 and 3.4 respectively. The pH of their salt (BA) solution is
Concept : This question is based on pH of WBWA salt solution.
Solution :

$$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$= 7 + \left(\frac{1}{2} \times 3.2\right) - \left(\frac{1}{2} \times 3.4\right) = 6.9$$

Final Answer : (4)

71)

Explaining : The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at

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T(K). The correct option for the pH of dimethylammonium acetate solution is.
Concept : This question is based on pH of WBWA salt solution.
Solution :

$$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$7 + \left(\frac{1}{2} \times 4.77\right) - \left(\frac{1}{2} \times 3.27\right) = 7.75$$

Final Answer : (3)

72)

Explaining : Match the correct pH of solution.
Concept : This question is based on pH of different solutions.
Solution :
(A) $pK_a + pk_b = 14$ for acid-base conjugate pair $pK_b = 14 - 6 = 8$
(B) $[H^+]_t = 10^{-8} + 10^{-7} = 11 \times 10^{-8}$; pH = 8–log11 = 8–1.04 = 6.96
(C) $pH = \frac{1}{2}pK_a - \frac{1}{2}\log C \Rightarrow pH = \left(\frac{1}{2} \times 4.7\right) - \frac{1}{2}\log 10^{-2} = 3.35$
(D) $[H^+] = \frac{10^{-3} \times 10^{-5}}{2} = 0.5 \times 10^{-3}$; pH = 4 – log5 = 4 – 0.7 = 3.3
Final Answer : (3)

73)

Explaining : What will be the pH of pure water at 60°C ?
Concept : This question is based on explanation of water.
Solution : pH decreases with increase in temperature.
Final Answer : (3)

74)

Explaining : The K_a values of acetic acid, hypochlorous acid and formic acid are 1.74×10^{-5}, 3.0×10^{-8} and 1.8×10^{-4} respectively Which of the following orders of pH of 0.1 mol dm^{-3} solutions of these acids is correct ?
Concept : This question is based on pH of weak acid.
Solution : Higher the value of K_a, Higher the acidic strength.
Final Answer : (2)

75)

Explaining : K_a for CH_3COOH is 1.8×10^{-5} and K_b for NH_4OH is 1.8×10^{-5}. The pH of ammonium acetate will be
Concept : This question is based on hydrolysis of WBWA salt.
Solution :

$$pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$\Rightarrow 7 + \left(\frac{1}{2} \times 4.74\right) - \left(\frac{1}{2} \times 4.74\right)$$

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Final Answer : (3)

76)

Explaining : Identify correct statement.
Concept : This question is based on common ion effect.
Solution : In presence of HCl, dissociation of H_2S decreases.
Final Answer : (2)

77)

Explaining : Identify the correct statement.
Concept : This question is based on hydrolysis of WBWA salt.
Solution : $pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
Final Answer : (1)

78)

Explaining : At 25°C, the dissociation constant of a base, BOH is 1.0×10^{-12}. The concentration of hydroxyl ions in 0.01 M aqueous solution of the base would be:
Concept : This question is based on Ostwald's dilution law.
Solution : $[OH^-] = \sqrt{K_b \times C} = \sqrt{10^{-12} \times 10^{-2}} = 10^{-7}$
Final Answer : (2)

79)

Explaining : Aqueous solution of which of the following compounds is the best conductor of electric current ?
Concept : This question is based on electrolytes.
Solution : Strong electrolytes are good conductors.
Final Answer : (4)

80)

Explaining : What is the conjugate base of OH^- ?
Concept : This question is based on acid-base conjugate pair.
Solution : $OH^- - H^+ \longrightarrow O^{-2}$
Final Answer : (4)

81)

Explaining : pH value of which of the following is NOT equal to one?
Concept : This question is based on pH calculation.
Solution :
(1) pH < 1 {W.A.}
(2) pH = $-\log 10^{-1} = 1$

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(3) pH = –log(0.05 × 2) = 1

(4) $[OH^+] = \frac{(0.4 \times 50) - (0.2 \times 5.0)}{100}$

$= \frac{10}{100} = 10^{-1}$

pH = –log 10^{-1} = 1

Final Answer : (1)

82)

Explaining : An acid HA ionises as $HA \rightleftharpoons H^+ + A^-$

The pH of 1.0 M solution is 5. Its dissociation constant would be

Concept : This question is based on pH of weak acid.

Solution :

$pH = \frac{1}{2}pK_a - \frac{1}{2}\log C$

$pK_a = 2 \times 5 = 10$

$K_a = 10^{-10}$

Final Answer : (1)

83)

Explaining : The pH of a 0.1 M aqueous solution of a weak acid (HA) is 3. What is its degree of dissociation?

Concept : This question is based on pH of weak acid.

Solution : $[H^+] = C.\alpha \Rightarrow 10^{-3} = 0.1 \propto \alpha \Rightarrow \alpha = 10^{-2} = 1\%$

Final Answer : (1)

84)

Explaining : Graph based question

Concept : This question is based on order of reaction.

Solution :

$-\frac{d[A]}{dt} = k[A]^1$; First order reaction

$A_t = A_0 - kt$; Zero order reaction

$\frac{1}{A_t} = \frac{1}{A_0} + kt$; Second order reaction

Final Answer : (4)

85)

Explaining : The acidic hydrolysis of ester is –

Concept : This question is based on pseudo 1st order reaction

Solution : Fact based

Final Answer : (1)

86) **Explaining :** For a first order reaction, how much half lives are nearly required for 99.9%

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completion of reaction? **Concept :** This question is based on first order reaction **Solution :** $t_{aq.g} = 10 \times t_{1/2}$ **Final Answer :** (4)

87)

Explaining : x, y and z are:
Concept : This question is based on rate of reaction and stoichiometry
Solution :

$$-\frac{1}{x}\frac{d[A]}{dt} = \frac{1}{y}\frac{d[B]}{dt} = +\frac{1}{z}\frac{d[B]}{dt}$$

$$\Rightarrow x : y : z :: 1 : 1 : \frac{2}{3}$$

Final Answer : (3)

88)

Explaining : Rate law calculation
Concept : This question is based on rate law calculation
Solution :
Keeping [A] constant when [B] is halved, rate becomes half.
= Order w.r.t B = 1 when [A] and [B] both are doubled, rate becomes 4 times
= Order w.r.t. A = 1
Final Answer : (2)

89)

Explaining : Calculation of order of reaction
Concept : This question is based on calculation of order of reaction
Solution :
Considering equation (i) and (ii), doubling [A], rate becomes 4 times, so order w.r.t [A] is 2.
from equation (i) and (iii), [B] double then rate also doubles order w.r.t B = 1.
Final Answer : (2)

90)

Explaining : Identify the correct statement
Concept : This question is based on Arrhenius equation
Solution : $k = A.e^{-Ea/RT}$
Final Answer : (4)

BIOLOGY

91) **Solution:** In water hyacinth and water lily, pollination performed by wind and insects
Final Answer: (3)

92) **Solution :** Stamen is male reproductive unit and outer to carpes.

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Final Answer : (1)

93) **Solution :** Mango, Apple, Guava, Lemon, Orange
Final Answer : (4)

94) **Solution :** The pollen tube enters the embryo sac through the micropyle and penetrates one of the synergid cells to release the male gametes.
Final Answer ; (1)

95) **Question Explanation**: The question asks the name of the process in which one male gamete moves towards the egg cell and fuses with it.
Concept: Fertilization
Solution:

A. Syngamy → fusion of male gamete with egg cell
B. Triple fusion → fusion of male gamete with two polar nuclei
C. Double fertilization → syngamy + triple fusion (in angiosperms)
D. Autogamy → self-pollination

Final answer: (1)

96) **Solution :** Option (B) and (C) are correct and option (A) and (D) are incorrect.
Final Answer : (3)

97) **Solution :** Zygote → Pro-embryo → Globular embryo → Heart shaped embryo→ Mature embryo
Final Answer : (2)

98) **Solution :** Papaya is dioecious.
Final Answer : (2)

99) **Solution :** Both statement I and statement II are correct
Final Answer : (1)

100) **Solution :** Option (3) incorrect. Bees are most dominant pollinator.
Final Answer : (3)

101) **Solution :** Flowers remains closed, thus pollen from other plant or flowers cannot pollenate these flower.
Final Answer : (1)

102) **Solution :** Secondary nucleus fusion of two polar nuclei (n + n)
Final Answer : (3)

103) **Solution :** 7 celled, 8 nucleus, 3 antipodal cells, 2 central cell, 2 synergids, 1 egg cell

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Final Answer : (3)

104) **Solution :** Option (4) incorrect both of them are pollinated by water
Final Answer : (4)

105) **Solution :** Option (3) as pollen grains are dry and non-sticky that can be carried by wind.
Final Answer : (3)

106) **Solution :** (b) and (d) are incorrect. pollen grain 25-50 μm outer 3 wall layer help in dehiscence.
Final Answer : (4)

107) **Solution :** In double fertilization:

A. One male gamete (n) + egg cell (n) → **Zygote (2n)**
B. Other male gamete (n) + two polar nuclei (n + n) → **Endosperm (3n)**

So, the ploidy is **Diploid (2n)** for zygote and **Triploid (3n)** for endosperm.
Final Answer : (2)

108) **Solution :** (d) incorrect : Part of embryonal axis just below level of scutellum is called hypocotyl.
(c) incorrect : rest zygotes divide only after certain amount of endosperm is formed.
Final Answer : (4)

109) **Solution :**
A - Antipodal
B - Polar nuclei
C- Synergids
D - Egg cell
Final Answer : (2)

110) **Solution :** Show the diagram in sequence
Final Answer : (3)

111) **Solution :**
Option (1) is incorrect : Viability varies in members of family also
Option (2) is incorrect : Both temperature and humidity affect viability of pollen.
Option (4) is incorrect : Rice and wheat pollen viability is about 30 mins.
Final Answer : (3)

112) **Solution :**

A.	Scutellum	II	Cotyledon of Monocot seed
B.	Non-albuminous seed	III.	Groundnut
C.	Epiblast	IV.	Rudimentary cotyledon

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D.	Perisperm	I.	Persistent nucellus

Final Answer : (1)

113) **Question Explanation:** We need to correctly match the pollen grain parts/structures with their features.

Concept: Pollen grain – structure

Solution: Exine → Outer layer, made of sporopollenin (very resistant).
Intine → Inner wall, made of pecto-cellulose.
Germ pore → Region where sporopollenin is absent (for pollen tube emergence).
Vegetative cell → Large cell with large irregular nucleus and abundant cytoplasm.

Final answer: Option 3: a–iii, b–iv, c–ii, d–i

114) **Solution :**

Column I		**Column II**	
A.	Funicle	V.	Stalk of ovule
B.	Hilum	IV.	Region where body of ovule fuses with funicle
C.	Integument	III.	One or two protective layers of ovule
D.	Chalaza	II.	Basal part of ovule
E.	Nucellus	I.	Mass of cells within ovule with more food

Final Answer : (2)

115) **Solution :** Endosperm in castor is not completely consumer and is available in nature seed.
Final Answer : (4)

116) **Solution :** In monoecious plants since male and female flowers are present on same plant so autogamy is not possible but geitonogamy and xenogamy is possible.
Final Answer : (1)

117) **Solution :** Option (2) incorrect the outer integument develop into testa, outer to seed coat layer
Final Answer : (2)

118) **Solution :** Embryonal axis above the level of cotyledons is epicotyl.
Final Answer : (2)

119) **Solution :** Option (4) incorrect endosperm development precedes embryo development

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Final Answer : (4)

120) **Solution :** The pollen grain represents the male gametophyte.
Final Answer : (3)

121) **Solution :** Leydig cells secrete testosterone under LH stimulation. Testosterone is essential for spermatogenesis (via Sertoli cells) and male secondary characters. Dysfunction directly collapses androgen-dependent processes.
Final Answer ; (2)

122) **Solution :** FSH drives proliferation of granulosa cells and growth of primary to secondary to Graafian follicles. Without it, no mature follicle is ready for ovulation.
Final Answer ; (2)

123) **Solution :** Primary oocyte completes meiosis-I just before ovulation forming secondary oocyte arrested at metaphase-II. Meiosis-II completes only after sperm entry to avoid wastage of resources if no fertilisation occurs.
Final Answer ; (2)

124) **Solution :** Acrosome contains hydrolytic enzymes (acrosin, hyaluronidase) released during acrosome reaction triggered by zona pellucida contact. Without it, sperm cannot digest the barriers.
Final Answer ; (1)

125) **Solution :** hCG mimics LH and sustains corpus luteum till placenta takes over progesterone production around 10–12 weeks.
Final Answer ; (2)

126) **Solution :** Trophoblast erodes endometrium and forms chorionic villi that interdigitate with uterine tissue to form placenta for exchange.
Final Answer ; (2)

127) **Solution :** Suckling stimulates posterior pituitary to release oxytocin, which contracts myoepithelial cells around alveoli.
Final Answer ; (2)

128) **Solution :** Without hCG, LH support falls; corpus luteum degenerates, progesterone crashes, leading to endometrial breakdown (menstruation).
Final Answer ; (1)

129) **Solution :** Zona pellucida glycoproteins (ZP3) bind sperm receptor and induce calcium influx leading to acrosome reaction.
Final Answer ; (1)

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130) **Solution :** Progesterone makes endometrium receptive (decidua) with increased vascularity and glycogen for implantation around day 21–24.
Final Answer ; (3)

131) **Solution :** Chorionic villi (fetal) + uterine decidua (maternal) = functional placenta for nutrient/gas exchange.
Final Answer ; (1)

132) **Solution :** Gastrulation establishes the three germ layers that give rise to all organs through differentiation.
Final Answer ; (1)

133) **Solution :** Foetal signals increase oxytocin sensitivity and release, causing positive feedback contractions
Final Answer ; (2)

134) **Solution :** Cleavage produces compact morula; (8-16 blastomeres) fluid accumulation later forms blastocoel in (32 or more blastomeres) blastocyst.
Final Answer ; (2)

135) **Solution :** Total two polar bodies are formed upto fertilization
Final Answer ; (2)

136) **Solution :** Sperm and ovum meet in ampulla; zygote is transported to uterus for implantation.
Final Answer ; (2)

137) **Solution :** hPL causes maternal insulin resistance and lipolysis to ensure foetal nutrition priority and prepare for locatation.
Final Answer ; (2)

138) **Solution :** This prevents wastage of cytoplasmic resources if no fertilisation occurs.
Final Answer ; (1)

139) **Solution :** Low progesterone withdraws support from endometrium leading to its shedding.
Final Answer ; (2)

140) **Solution :** Placenta separates; open cord can cause haemorrhage or infection.
Final Answer ; (1)

141) **Solution :** Continuous mitotic renewal of spermatogonia allows lifelong sperm production; oogenesis is limited and unequal cytoplasmic division conserves resources for one

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ovum.
Final Answer ; (3)

142) **Solution :** Cortical granules release enzymes that modify zona pellucida making it impenetrable to additional sperms.
Final Answer ; (3)

143) **Solution :** Placenta only during pregnancy and secretes many hormones.
Final Answer ; (3)

144) **Solution :** No G1/G2 growth phases in cleavage; compact morula/ blastocyst reaches uterus in 4–5 days.
Final Answer ; (3)

145) **Solution :** Yolk sac is vestigial in humans (no yolk); other three are true.
Final Answer ; (1)

146) **Solution :** Menstruation is caused by fall in progesterone.
Final Answer ; (1)

147) **Solution :** High steroid hormones from placenta exert negative feedback on hypothalamus and pituitary preventing new cycle.
Final Answer ; (1)

148) **Solution :** Placenta forms only during pregnancy and secrete many hormones.
Final Answer ; (1)

149) **Solution :** Each structure performs specialised supportive or secretory roles essential for gamete maturation and pregnancy maintenance.
Final Answer ; (2)

150) **Solution :** Hormonal peaks and withdrawal precisely time ovulation, pregnancy support, birth and cycle reset.
Final Answer ; (1)

151) **Solution :** ICSI injects a single selected sperm directly into the egg cytoplasm in a lab, overcoming low count/motility issues that other methods cannot. It enables fertilization even with minimal viable sperm while keeping the process under controlled conditions for higher success.
Final Answer ; (2)

152) **Solution :** The condom forms a physical barrier to both sperm and pathogens, providing dual protection; other methods only prevent pregnancy and leave STD risk unaddressed.

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Final Answer ; (3)

153) **Solution :** Second-trimester MTP involves larger fetal size and more complex evacuation, increasing bleeding, infection, and perforation risks compared to simpler first-trimester methods.
Final Answer ; (2)

154) **Solution :** It balances the right to healthy offspring detection against misuse that skews sex ratio, upholding reproductive health as social and ethical well-being.
Final Answer ; (2)

155) **Solution :** IUT transfers the in-vitro fertilized embryo/zygote into the uterus, bypassing blocked tubes entirely.
Final Answer ; (2)

156) **Solution :** It relies on avoiding coitus during the fertile window (ovulation ± few days); any miscalculation in cycle prediction leads to failure, unlike barrier or hormonal methods.
Final Answer ; (1)

157) **Solution :** Exclusive breastfeeding suppresses ovulation for ~6 months via prolactin, offering free, natural spacing without devices or surgery.
Final Answer ; (2)

158) **Solution :** The vas deferens is cut and tied, blocking sperm transport; seminal fluid from prostate/seminal vesicles remains, so ejaculation feels normal and hormones are unaffected.
Final Answer ; (1)

159) **Solution :** They suppress FSH/LH (no ovulation), thicken cervical mucus (sperm block), and make endometrium unsuitable for implantation — triple protection.
Final Answer ; (2)

160) **Solution :** It expanded from birth control to holistic care including awareness, maternal health, STD treatment, and infertility support.
Final Answer ; (2)

161) **Solution :**They delay or prevent ovulation/implantation if taken early; efficacy drops sharply after 72 hours.
Final Answer ; (1)

162) **Solution :**Hormonal and IUD methods do not block viral entry; only barriers like condoms do.
Final Answer ; (4)

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163) **Solution :**Prolactin surge during exclusive breastfeeding reliably delays ovulation for 6 months.
Final Answer ; (3)

164) **Solution :**Once-weekly intake reduces forgetfulness and maintains normal cycles better than daily hormonal pills
Final Answer ; (2)

165) **Solution :**Holistic RCH integrates counseling, free contraceptives, safe MTP, and child/maternal care, leading to voluntary small families.
Final Answer ; (2)

166) **Solution :**The RCH program aims to improve maternal and child health, reduce infant mortality, and promote reproductive health.
Final Answer ; (2)

167) **Solution :**Oral contraceptive pills contain hormones that suppress ovulation.
Final Answer ; (3)

168) **Solution :**Vasectomy involves cutting and sealing the vas deferens to prevent sperm release.
Final Answer ; (2)

169) **Solution :**HIV/AIDS is caused by the Human Immunodeficiency Virus, while others listed are bacterial infections.
Final Answer ; (3)

170) **Solution :**Amniocentesis is a prenatal diagnostic technique to detect chromosomal abnormalities in the fetus.
Final Answer ; (2)

171) **Solution :**Coitus interruptus (withdrawal method) is a natural contraception method, unlike others which are artificial.
Final Answer ; (2)

172) **Solution :**Copper ions create a spermicidal environment and alter endometrium; however, it is an IUD (intra-uterine device), not a barrier method.
Final Answer ; (3)

173) **Solution :**Second-trimester MTP is riskier; first-trimester (up to 12 weeks) needs only one doctor’s opinion as per MTP Act.
Final Answer ; (4)

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174) **Solution :**ART includes IVF, ICSI, etc., done with the biological mother; surrogacy is only one option when the mother cannot carry the pregnancy.
Final Answer ; (3)

175) **Solution :**STDs spread via sexual contact; viral ones like HIV/Hep-B have no complete cure; early intervention prevents long-term damage; condoms reduce but do not eliminate all risks (e.g., skin-to-skin).
Final Answer ; (2)

176) **Solution :**Both are highly effective and permanent; reversal is difficult and not guaranteed; hormones and libido remain normal.
Final Answer ; (1)

177) **Solution :**They need 7 days for full effect; they do not increase STD risk but offer no protection
Final Answer ; (1)

178) **Solution :**Barrier action is the key mechanism.
Final Answer ; (1)

179) **Solution :** Infertility can be male, female, or combined.
Final Answer : (3)

180) **Solution :** Each method targets a different reproductive step, allowing choice based on individual needs and health status.
Final Answer : (4)

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